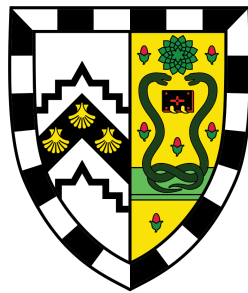


Analysis of Low Cost Particle Image Velocimetry

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I hereby declare that, except where specifically indicated, the work submitted herein is
my own original work

Technical Abstract

Particle image velocimetry (PIV) is a method used in experimental fluid mechanics to measure instantaneous velocity vector fields. A conventional planar PIV system seeds the fluid with numerous fine particles, illuminates these particles with a thin laser light sheet, and captures two consecutive images of the particles spaced by the interframing time. Spatial image cross correlation is performed to determine the average displacement of particles in distinct ‘interrogation spots’, and the interframing time is used to calculate the measured velocity vectors.

This report begins by introducing and motivating the project and then reviewing existing literature on PIV and its limitations, low cost PIV, and existing PIV analysis tools. A literature review finds that no tools currently exist capable of performing all of the following tasks: synthetic PIV image generation including pixelisation, particle simulation and optical modelling; velocity estimation from PIV images; and comparison of simulated velocity measurements to an exact underlying flow. A tool called the digital twin is developed to fill this gap. This enables the errors associated with any PIV system to be quantified before system construction.

The digital twin can quantify and characterise PIV system errors. Dimensional analysis is used to find the twelve dimensionless groups on which median dimensionless PIV system error depends, enabling generalised conclusions about the effects of hardware parameters on system errors to be made. Three of these dimensionless groups are shown not to affect dimensionless error. The errors associated with the remaining nine dimensionless groups are characterised and numerical bounds in which errors are small for a ‘typical’ system are found.

Four error categories are found using the digital twin: pixelization errors, false correlation errors, particle slip errors, and velocity vector location assignment errors.

Quantisation or pixelization errors are errors due to the system’s inability to determine particle locations with a resolution greater than half a pixel, resulting in quantised velocity vector measurements. The magnitude of the normalised quantisation error is affected by the dimensionless resolution, a dimensionless characteristic particle travel distance, and system geometry parameters.

False correlation errors are errors due to the correlation algorithm finding a region of the second image which produces a higher correlation peak than the region corresponding to the group of particles being tracked between the two images. These errors are always large. Their frequency is affected by many dimensionless parameters, but particularly important properties are those affecting the magnitude of flow shear, the size of the interrogation spot used to track particles between images, and the maximum distance travelled by particles during the interframing time.

A particle needs to travel at a velocity different to that of the surrounding flow in order

to generate a drag force. Therefore particles in regions of accelerating flow must ‘slip’ in order to generate the force required for the particle to follow the flow. This relative slip velocity causes particle slip errors due to the particles not perfectly following the flow. For a fixed choice of the underlying flow field particle slip errors are governed by Stokes number only.

The average velocity of particles within an interrogation spot will be representative of the flow velocity at some point within the interrogation region, but this point is not necessarily the centre of the first interrogation spot. This means that assigning the measured velocity vector to the centre of the first interrogation spot introduces an error, known as a velocity vector location assignment error. These errors are primarily affected by the dimensionless parameter governing the size of an interrogation spot.

A hardware review reveals that pulsed lasers offer instantaneous luminous power ratings five orders of magnitude greater than off the shelf LEDs while LEDs are around three orders of magnitude cheaper. Specialised CCD cameras are more expensive than most CMOS cameras, but CCD cameras offer interframing times around three orders of magnitude smaller than CMOS cameras, leading to better system performance.

Most importantly, the usefulness of the digital twin as a tool for characterising PIV system errors and providing quantitative error magnitude data for candidate PIV system designs is demonstrated in three case studies. The digital twin is used to assist the conceptual design of a water based PIV system and a novel dual-camera low cost gaseous PIV system. The digital twin is also used to show how increasing air pressure and Reynolds matching can decrease experimental uncertainty by reducing particle slip errors.

Future work includes extending the digital twin to link the illumination model to real hardware parameters and to model out of plane particle motion effects. Additionally, the low cost PIV systems and experimental methods proposed in the case studies should be tested. Finally, implementing different underlying flow fields and correlation methods with sub-pixel accuracy would improve the digital twin’s ability to model real PIV measurements made by state of the art systems.

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1 Introduction

Traditional velocity probes include pitot tubes, hot wire anemometers, and laser doppler velocimetry systems. These probes measure velocity at a single point and hence visualising instantaneous flow structures and quantifying vorticity is difficult. Traditional flow visualisation techniques, such as using smoke, dye injection, or hydrogen bubbles, enable the visualisation of instantaneous flow structures but do not provide quantitative data. PIV is useful because it measures the entire velocity field simultaneously, enabling the measurement of instantaneous flow structures and the vorticity field. PIV usage in fluid dynamics research is limited by cost. Conventional PIV systems cost upwards of £50,000 [2], providing a barrier to entry for low-budget projects and teaching applications.

In this project a tool called the digital twin is developed to quantify and characterise the errors associated with a given PIV system, enabling designers to choose hardware based on simulated measurement error magnitudes rather than unreliable heuristics. Dimensional analysis is used to find the dimensionless groups on which the median dimensionless error depends, and the digital twin is used to analyse the sensitivity of a ‘typical’ system’s dimensionless error to variations in these dimensionless parameters. A hardware review is performed, and finally three case studies are presented to demonstrate the usefulness of the digital twin as both a tool for system design (with a focus on cost reduction) and a tool for existing system optimisation and error characterisation.

2 Literature Review

2.1 PIV Fundamentals

Conventional planar PIV uses double-pulsed lasers producing light pulses with the energies required to create visible images of micrometer sized tracer particles (between 5 and 500 mJ [2]) at a wavelength of 532 nm. The laser pulse duration is usually much shorter than the camera frame exposure so the laser pulse duration dictates the image exposure time.

Seeding particles must be large enough to scatter sufficient light energy to form bright images, but small enough that they accurately follow the fluid acceleration. They typically have diameters on the order of micrometres for gaseous flows. The particles must be monodisperse (uniform in diameter) to high accuracy to avoid variations in scattered light energy between particles, as cameras with automatic exposures will only capture the brightest particle images and therefore variations in particle diameter lead to an effective reduction in tracer particle number density.[2] Particles must also be spherical to avoid variations in scattered light energy due to particle rotation, as this causes errors in displacement estimation.[2] Common PIV tracer particles include thermoplastic microspheres, hollow glass microspheres, polystyrene spheres, and atomised oil droplets.

A typical planar PIV system uses a single lens to focus scattered light onto a charge-coupled device (CCD) video chip, which converts the light energy into a charge output. Double-shutter CCD cameras have two sets of adjacent electron storage wells, allowing a captured frame to be transferred to the adjacent electron storage wells in around 1 μ s. This enables two frames to be captured in rapid succession. The short interframing times achievable with CCD cameras make them the preferred choice for PIV. CMOS cameras are a cheaper option, but long interframing times (on the order of milliseconds due to the time taken to transfer data from the frame grabber to RAM) restrict their performance.

After image capture, the images are divided into equally sized ‘windows’ and spatial image cross correlation is used to estimate the mean displacement of particles in each window. The mean displacements and the interframing time are then used to calculate the velocity field. This process is described in detail in section 3.6.

Other types of PIV include time resolved PIV (TR-PIV), statistical PIV, and tomographic PIV. TR-PIV uses high framerate cameras to measure time-varying phenomena such as the onset of turbulence. Current state of the art TR-PIV systems cannot measure Kolmogorov scale fluctuations but are capable of resolving flow at the Taylor microscale.[13] Statistical PIV uses high resolution cameras to take an ensemble of PIV measurements in order to measure flow statistics such as mean velocities or Reynolds stress tensor components. TR-PIV and statistical PIV systems have different design aims but can both be achieved using a planar PIV system. Finally, tomographic PIV illuminates a measurement volume and used multiple cameras to reconstruct the 3D flow field within this volume.

State of the art PIV systems can typically measure velocity at between 10^3 and 10^5 positions and yield velocity measurements with an accuracy of 1 to 2% of the maximum measurable velocity[13]. Other methods of flow measurement by particle tracking have been investigated, such as particle streak velocimetry (where a single exposure is used to create particle streaks from which velocity is inferred), laser speckle velocimetry (where the ‘speckle’ caused by many overlapping particles is tracked, permitting very high particle seeding number density) and particle tracking velocimetry (where individual particles are tracked, limiting the spatial density of velocity vector measurements), but it was concluded that “double-pulsed, high-image-density images analysed by digital spatial correlation analysis”[1] is the most promising particle-imaging technique for experimental fluid-dynamics.

State of the art PIV system accuracy is limited by particle slip which causes errors on the order of 1%, errors due to the finite light sheet thickness and magnification uncertainty on the order of 0.5%, particle image blurring, pixelisation, and uncertain data processing techniques.[13] Pixelisation, particle slip and image blur are especially important when attempting to reduce cost by compromising on hardware specifications.

2.2 Low Cost PIV

Various attempts to reduce the cost of PIV have been made. Acquah et al.[4] built a low-cost (\$520.5) PIV system in 2024 designed to measure the flow of water in a stirred beaker. The system used an off the shelf camera, laser module, and optics, and used hollow ceramic spheres as its tracer particles. It achieved an accuracy of “between 1–2% when the flow stayed two-dimensional”. [4]

Ring and Lemley[10] built a low-cost (\$609, excluding software costs) PIV system to measure the flow of a water and glycerin mixture for a “Reynolds number range of approximately one to 11” [10], achieving a fluid velocity measurement accuracy of “almost seven percent”. Their approach used an off the shelf laser pointer, camera, and optics kit, and used acrylic shavings as tracer particles.

Many other low-cost PIV systems for measuring water flows have been developed [11][14], but low-cost PIV in air at Reynolds numbers greater than 10^3 has not been achieved. The difficulty in designing a low-cost gaseous PIV system is that the tracer particles must be very small to accurately follow the flow, whereas in a liquid PIV system particles with diameters in the 10 μm range can be used with acceptable slip error magnitudes by matching the tracer particle density to the fluid density.[14] The small particles required for gaseous PIV typically require 5 to 500 mJ light pulses to create visible particle images, requiring the use of expensive pulsed laser sources.

Nader et al.[5] achieved low-cost PIV measurements of the wake produced by a 10 mm diameter cylinder in air with a mean velocity of 0.2 m/s. This corresponds to a Reynolds number of approximately 130. However, the results were “very noisy” [5] and “further development is necessary”. [5]

2.3 PIV Analysis Tools

Two existing tools which provide some of the functionality required to characterise the errors of a PIV system and aid with PIV system design are PIVLAB[8] and PIV-Image-Generator[7]. PIVLAB is “a free and open-source particle image velocimetry (PIV) software and is currently the most frequently cited PIV tool on the market” [8] “It can be used to calculate the velocity distribution within imported (or captured) images” [8] and provides other useful analysis functionalities. However, it cannot generate synthetic PIV images and therefore cannot be used to assess the errors present in a hypothetical PIV system.

PIV-Image-Generator[7] generates “synthetic images of tracers in turbulent flows to benchmark Particle Image Velocimetry and Optical Flow algorithms” [7]. This is useful for the stated purpose of benchmarking PIV algorithms, but the assumption is made that the tracer particles “are small and ideally have a density close to that of the medium, so that their movement is highly related with the flow or the medium.” [7] This assumption

causes the omission of errors due to particle drift which are critical for designing a low-cost PIV system for use in gaseous flows.

The authors of PIV-Image-Generator found that there were no existing tools with all the functionality necessary to benchmark PIV algorithms.[7] To the best of my knowledge, there are still no tools capable of performing all the following tasks: synthetic generation of representative PIV images including particle slip errors, pixelisation, and an optical model; velocity estimation using synthetic or real PIV images; and comparison of a generated velocity field to an exact known velocity field. The lack of such a tool hinders the development of low-cost PIV systems for gaseous flows. This report bridges this gap through the development of such a tool.

2.4 Research Questions

The prior introduction and literature review leads to three research questions:

1. What are the sensitivities of 2D PIV measurement errors to variations in hardware properties?
2. How do hardware properties scale with cost?
3. What hardware changes offer the greatest potential cost reduction for planar PIV in gaseous flows?

Research question 1 leads to the development of a tool capable of synthetic image generation, velocity estimation, and error characterisation for a hypothetical PIV system. This tool, henceforth termed the ‘digital twin’, is used to analyse the sensitivity of a PIV system’s median dimensionless error to variations in dimensionless system parameters, answering research question 1.

Research question 2 is addressed by a hardware review. Research question 3 is answered by combining the results of research questions 1 and 2 to suggest some novel hardware choices and experimental techniques in three case studies, demonstrating the usefulness of the digital twin and providing a starting point for future experimental work.

3 Digital Twin

3.1 Motivation

Answering research question 1 requires either comprehensive testing of different planar PIV setups, or computational simulation of planar PIV systems with modifiable parameters. Large scale testing would be unjustifiably expensive and time consuming and is therefore not a viable approach to determining how PIV errors scale with system parameters. Instead, the choice was made to develop the ‘Digital Twin’, a simulated model of a

2D PIV system designed to characterise the errors of a potential low cost 2D PIV system, answering research question 1. The PIV Digital Twin enables the quantitative evaluation of the performance of different PIV systems, enabling assessment of the viability of low cost PIV systems. This assists in answering research question 3.

(N.B. No existing software is capable of synthetic PIV image generation including particle slip errors, velocity estimation, and calculation of velocity estimation errors relative to a known velocity field. Therefore no existing software is capable of answering research question 1. Section 2.3 explores in detail existing software and its shortcomings.)

The digital twin is programmed in Python. Python was chosen for its myriad of well developed scientific libraries, ease of programming, and being open source.

3.2 Architecture

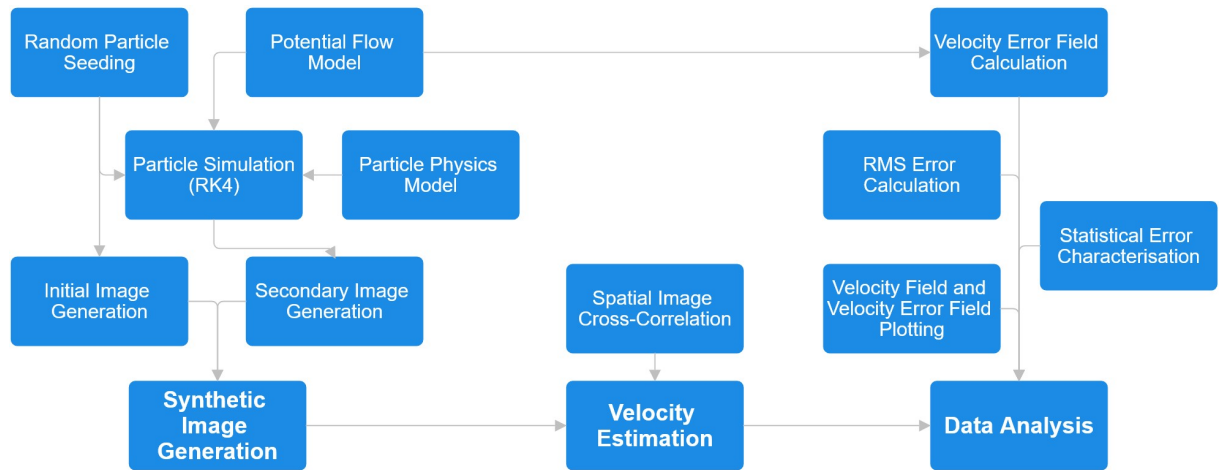


Figure 1: Digital Twin Architecture

The digital twin is designed to quantify and characterise the measurement errors present in any PIV system, given only the system and flow field parameters. To this end, it must perform three primary functions: synthetic image generation, velocity estimation, and data analysis.

Synthetic image generation is the process of computationally generating images representative of what a given PIV system may produce. To achieve this, an underlying flow field is prescribed and particles are randomly ‘seeded’ in the flow. The particle motion is then simulated during the first exposure, utilising a particle physics model based on Stokes flow and fourth order Runge-Kutta numerical integration. The particle traces, an illumination model, and pixelisation based on camera specifications are then used to generate the first image. The motion of the particles during the interframing time is then simulated, and a second image is generated using the same process as for the first image. Both images can then be saved as outputs and used in the subsequent analysis.

Velocity estimation is the purpose of real PIV systems. To characterise the measurement errors inherent in a given PIV system, we must perform velocity estimation to generate a representative velocity field measurement, and then compare this velocity estimate to the true flow field velocity to determine the errors. Velocity estimation is performed by spatial image cross correlation, a process which is described in detail in section 3.6.

Finally, data analysis is required to gain useful design information from the digital twin. The calculation of a velocity error field is performed by subtracting the true flow velocity from the estimated flow velocity for all estimated velocity vectors. RMS error calculation and other statistical error characterisations can then be performed, enabling insight into both the magnitude and cause of measurement errors. Finally, a suite of functions to plot velocity fields, error magnitude histograms, sensitivity graphs, and more is provided enabling the generated data to be inspected easily.

3.3 Fluid Simulation

The purpose of the digital twin is to characterise and quantify the expected errors intrinsic in a given PIV system design. The purpose is not to simulate real fluid flows, although the tool is easily modifiable to work with any prescribed flow field. Potential flow around a cylinder was chosen as a sensible and easy to implement underlying flow field.

Potential flow is simple and useful as it can be solved analytically. Flow accuracy does not matter as we are concerned with how well a given PIV system reconstructs the flow, not how accurate the flow model is. Hence the unphysical, inviscid nature of potential flow is not an issue for the digital twin.

Equation (1) defines the velocity field for potential flow around an infinite cylinder, where ϕ is the velocity potential, U is the freestream velocity, a is the cylinder radius, $\mathbf{u}$ is the velocity of the potential flow, and x , r and θ are coordinates with the usual definitions. Equation (1) has an exact solution for $\mathbf{u}$, given by equation (2). This exact solution is the underlying flow model implemented.

$$\phi = Ux - \frac{\alpha \cos \theta}{2\pi r}, \quad \alpha = -2\pi a^2 U, \quad \mathbf{u} = \nabla \phi \quad (1)$$

$$u_r = U(1 - a/r)^2 \cos(\theta) \quad u_\theta = -U(1 + a/r)^2 \sin(\theta) \quad (2)$$

3.4 Particle Simulation

PIV is highly susceptible to particle slip errors, so it cannot be assumed that the particles follow the flow perfectly. “In most airflows seeded with small 1-2 μm droplets or particles, the fidelity with which tracer particles follow the fluid motion is somewhat less than

1%.”[13] Therefore a particle physics model is required. The equation of motion for a small, spherical particle at low Reynolds number, immersed in a fluid flow is given by equation (3).[2] The definition of τ_p , a characteristic time constant associated with relative velocity decay, shows that increasing particle diameter or density will result in a larger characteristic time constant and thus a particle which ‘slips’ more.

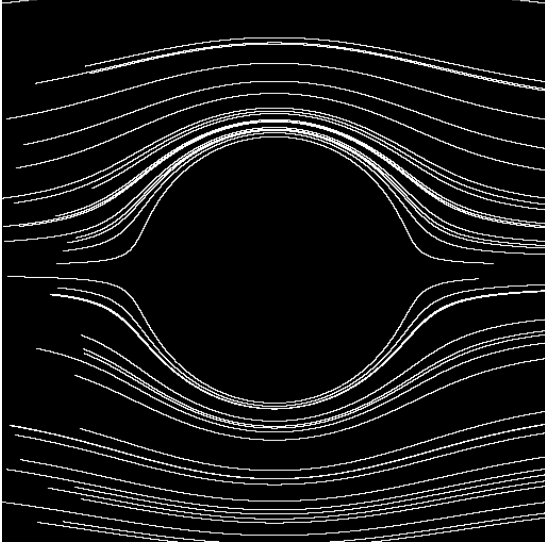
The 4th order Runge-Kutta (RK4) numerical method [3] is used to simulate the particles. A 4D ‘state vector’ containing position and velocity information is defined for all particles, and RK4 is used to update this state vector with new position and velocity information after each time step. Equation (4) shows the RK4 numerical method for updating a state vector $\mathbf{z}$ governed by the equation $\dot{\mathbf{z}} = f(t, \mathbf{z})$. The RK4 numerical method is chosen due to its fourth order accuracy making it appropriate for scientific applications.

$$\dot{\mathbf{v}}_p \approx \frac{\mathbf{u} - \mathbf{v}_p}{\tau_p} + \mathbf{b}, \quad \tau_p = \frac{(\rho_p - \rho_f)d_p^2}{18\rho_f\nu_f\phi}, \quad \phi \approx 1 \text{ for } Re \ll 1 \quad (3)$$

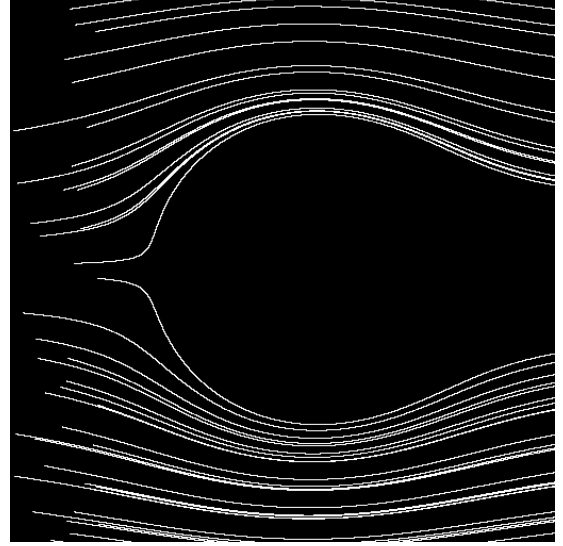
$$\begin{aligned} \dot{\mathbf{z}} &= f(t, \mathbf{z}) \\ \mathbf{z}(t_{n+1}) &= \mathbf{z}(t_n) + \frac{1}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)\Delta t + \mathcal{O}(\Delta t^5) \\ \mathbf{k}_1 &= f(t_n, \mathbf{z}(t_n)) \\ \mathbf{k}_2 &= f(t_n + \Delta t/2, \mathbf{z}(t_n) + \mathbf{k}_1\Delta t/2) \\ \mathbf{k}_3 &= f(t_n + \Delta t/2, \mathbf{z}(t_n) + \mathbf{k}_2\Delta t/2) \\ \mathbf{k}_4 &= f(t_n + \Delta t, \mathbf{z}(t_n) + \mathbf{k}_3\Delta t) \end{aligned} \quad (4)$$

The choice of time step for RK4 is an important consideration. The characteristic time constant τ_p is very small for micrometer sized particles (on the order of tens of nanoseconds) and it is found that the simulation becomes unstable if $\Delta t > \tau_p e$, where $e \approx 2.718$ is Euler’s number. The simulation is found to be sufficiently accurate for $\Delta t < \tau_p$, and hence the condition $\Delta t < \tau_p/2$ is enforced for all simulations to ensure simulation stability and accuracy.

Figures (2a) and (2b) demonstrate the effect of particle slip using long exposure synthetic images. (Details of image generation are postponed until section 3.6.) Figure (2a) shows the traces of infinitesimal particles which follow the flow exactly, generated by changing the equation of motion to set particle velocities to the exact underlying flow velocity. It looks like the expected potential flow around a cylinder, providing a useful sanity check for the particle and flow simulation. Figure (2b) shows the traces of particles with large characteristic time constant. These particles visibly slip and produce a very different image to figure (2a), demonstrating the issue of large particles not following the flow and thus introducing errors in PIV measurements.



(a) Long exposure



(b) Long exposure - Slip

Figure 2: Synthetic images generated by the Digital Twin

3.5 Optical Model and Synthetic Image Generation

Diffraction causes point particles to generate finite-diameter images, and unless the imaging is microscopic or the particles are large, image diameter is almost completely dominated by diffraction.[2] Pixelisation limits the accuracy with which the centre of a small particle image can be determined to plus or minus half a pixel, leading to a phenomenon known as pixel locking.[2] Prasad[9] found that “particle image diameter strongly influences accuracy”[9] and that “the total error is minimized by selecting d_τ such that $d_\tau/d_{\text{pix}} \approx 2$ ”[9], where d_τ is the diameter of the particle image and d_{pix} is the pixel pitch in metres.

Clearly particle illumination is a key consideration in the design of a PIV system and it would be incorrect to assume that particles which are small relative to the pixel pitch create particle images which are small relative to a pixel. Thus a model of the optical behaviour of illuminated particles must be included in the digital twin.

The digital twin models the optical properties of illuminated small particles by assuming that the particles are monodisperse and spherical to high accuracy. Each particle can therefore be treated using the same illumination properties. The digital twin assumes the illumination of the final image created by the light energy scattered by a small particle and focused into a camera sensor can be modelled as a 2D Gaussian distribution. Hence the ‘brightness’ incident of the camera sensor from a single particle scattering light energy is instantaneously a ‘circular blob’ with ‘brightness’ variation proportional to equation (5).

$$\text{brightness} \propto \exp\left(-\frac{X^2 + Y^2}{2\sigma^2}\right) \quad (5)$$

The digital twin models the pixel illumination due to this incident brightness by as-

suming the particle is central within the pixel and assuming the total brightness of each pixel is equal to the Gaussian evaluated at the corresponding pixel centre. These assumptions allow the final image to be computed efficiently by first generating an image by looping through all particle locations and adding unity brightness to the pixel each particle instance lies in, then multiplying this image by the illumination intensity, convolving this image with a Gaussian filter with standard deviation equal to ‘particle image sigma’, and finally converting the image to int8 format, clipping any pixel brightnesses above 255. Images are displayed using the OpenCV library for python.

Figure (3) shows an example image generated for a 512×512 pixel² camera imaging the particles in a potential flow behind a cylinder using a moderate illumination intensity and particle sigma = 3 pixels. The cylinder lies just to the left of the image and the curvature of particle paths near the cylinder can be easily seen. The blur of the particles caused by the illumination model is clearly shown. Additionally, one particle halfway up the left hand side of the image is notably brighter than all other particles due to its lower velocity. (This fact lends itself to the idea of determining speed by particle brightness, but in practice particles are not perfectly monodisperse and hence this technique is inviable.) Brighter streaks are also seen where particles are near each other.

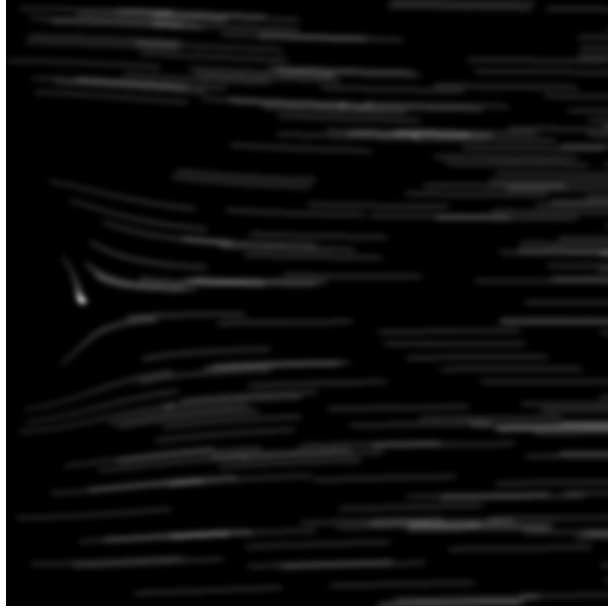


Figure 3: Synthetic long exposure image of tracer particles using a 512×512 pixel² camera

This model illumination model has some important limitations which are worth noting before making further progress. The first is that images are generated based on which pixel a particle lies at each time frame in rather than the exact position of the particle at each time frame. This causes a small illumination error. For example, if a particle sits very near the edge of one pixel then the adjacent pixel should have almost equal brightness, whereas this optical model will create brightness assuming the particle lies at the centre of the pixel. Additionally, pixel brightness due to a single particle at a single

time instance should be calculated by integration of the Gaussian brightness distribution over the domain of the pixel, but is instead approximated by assuming that the pixel brightness imparted is equal to the value of the Gaussian at the centre of the pixel. Both of these errors cause errors in illumination position of less than half a pixel. Leaving these error sources in the digital twin enables the use SciPy's [12] Gaussian filter convolution function. Generating images without using convolution, which is necessary to address these errors, leads to unacceptably slow image generation.

Another important limitation of the model is the assumption that particles are monodisperse and spherical to high accuracy. Real particles used for PIV vary in size and are not perfectly spherical, although it is preferable to the most monodisperse and spherical particles possible within the constraints imposed by particle slip and particle visibility. Real particles therefore create particle images of varying size. Extending the model to include particles with varying brightnesses is possible, but varying the Gaussian illumination distribution's standard deviation for difficult particles is not, again due to the unacceptably slow performance of image generation without using convolution. Dealing with non-spherical particles is also impractical for the same reason.

The illumination intensity and particle image diameter parameters in the digital twin are inputs which could have their values derived through a physical model. The choice has been made not to do this within the digital twin, as the benefit of directly relating particle, laser, camera and optics properties to particle image diameter and illumination intensity is outweighed by the time required to fully understand the complex subject of optics, and by the need for many more input parameters yielding a more complicated but not necessarily more informative sensitivity analysis. However, the basic theory to relate particle image diameter to other properties is outlined below to aid understanding of the design choices associated with PIV illumination and optics.

"The finite diameter of the particles, d_p , the effect of diffraction by the lens, and the effect of geometric lens aberrations all combine to cause the diameter of an image to have finite value given approximately by

$$d_\tau \approx (M_0^2 d_p^2 + d_s^2 + d_a^2)^{1/2}$$

where

$$d_s = 2.44(1 + M_0)f^\# \lambda$$

is the diffraction limited spot diameter,

$$f^\# = f/D_a$$

the f-number of the lens, D_a the diameter of the lens aperture, λ the wavelength of the light, and d_a the diameter of the aberrated image of a point source." [2] f is the lens focal

length and M_0 is the paraxial lateral magnification, defined as

$$M_0 = Z_0/z_0$$

where Z_0 and z_0 are the distances from the effective centre of the lens to the image plane and the object plane respectively. For a lens with constant magnification, the location of the centre of the image of a point particle can be found by drawing a straight line from the object point through the effective centre of the lens to the image plane.[2] Thus, assuming a single lens of constant lateral magnification is used, geometry shows

$$M_0 = W_0/w_0$$

where W_0 and w_0 are the camera sensor and field of view widths respectively.

“For a thin lens of focal length f , the condition for focusing points in the object plane ($z = 0$) into the image plane is the Gauss lens law.”[2]

$$\frac{1}{Z_0} + \frac{1}{z_0} = \frac{1}{f}.$$

In order to relate the particle image diameter to the illumination and optical parameters without delving into optics in excessive complexity, it is assumed that the image is in focus, seeding particles are perfectly monodisperse and spherical, and lenses are high quality such that lens aberrations are negligible. These assumptions and the prior equations lead to

$$d_\tau \approx \left(\frac{W_0^2}{w_0^2} d_p^2 + 2.44 \left(1 + \frac{W_0}{w_0} \right) \frac{Z_0 \lambda}{(1 + W_0/w_0) D_a} \right)^{1/2}$$

and thus if the particle diameter, system geometry, light wavelength, and lens aperture size is known then the particle image diameter in metres can be determined. This can then be multiplied by $\sqrt{R}/w_0$ to get the particle image diameter in pixels, as used in the digital twin.

The image intensity parameter is based on the illumination pulse energy, optical properties, and camera properties. The complexities of modelling a camera’s quantised response to incident light energy are best left to a student of optics.

3.6 Velocity Estimation

Velocity estimation is required to complete the simulation of a PIV velocity field measurement. The images are split up into interrogation regions or windows, typically 32×32 pixels² in size [2], and windows are often allowed to overlap to achieve higher spatial resolutions, though this yields measurements which are not independent. The cross-correlation

function is defined by equation (6), and is a measure of how similar two windows are when one is shifted by a shift vector $\mathbf{s}$. The idea of spatial image cross-correlation is to search for the shift vector which maximises cross-correlation for a given window, and assume this shift vector corresponds to the average displacement of the particles within that window. This concept is visually represented in figure (4). Once the shift vector which maximises cross correlation is found, the local flow velocity estimate is computed by equation (7).

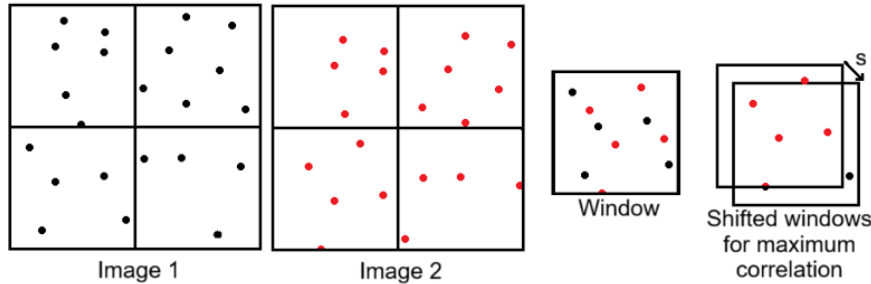


Figure 4: A visual representation of the spatial image cross-correlation process

$$R(s) = \int_{W_1} I_1(\mathbf{X}) I_2(\mathbf{X} - \mathbf{s}) d\mathbf{X} \quad (6)$$

$$\mathbf{v} = M_0 \frac{\mathbf{s}}{t} \quad (7)$$

A limit must be placed on the size of the shift vector, both to avoid false correlation errors with far away similar looking regions of particles, and to determine the shift vectors which maximise cross-correlation in a reasonable time frame. (Searching two 1024×1024 pixel² images split up into non-overlapping 32×32 pixel² interrogation regions for every possible shift vector would require computing the cross-correlations of around 1 billion distinct pairs of windows, taking an unreasonable amount of time.)

The Digital Twin's interrogation algorithm searches for shift vectors up to a maximum displacement, which is determined by the maximum measurable velocity, a parameter input by the user. For analysis performed later in the report, this is set equal to 2.2 times the potential flow free-stream velocity, ensuring that the whole velocity range seen in the potential flow is captured and some headroom exists for measurement errors. Once the cross-correlation maximising shift vectors have been found, equation (7) is used to calculate the average velocity of the interrogation region, which is assigned to the location of the centre of the first image's interrogation region. (See [2] for definitions of the symbols in equations (6) and (7).)

The spatial image cross-correlation method of velocity estimation is preferred to tracking individual particles, known as Particle Tracking Velocimetry (PTV), as greater spatial resolutions can be achieved [2]. PTV necessitates a low density of particles so that individual particles can be distinguished between images, limiting spatial resolution.

3.7 Error Calculation and Characterisation

Velocity measurement errors are calculated by the digital twin by subtracting the underlying flow velocity at the appropriate point from each velocity vector returned from the velocity estimation step, creating a vector field of velocity errors. Both the measured velocity vector field and the velocity error vector field can be plotted, assisting the user in determining the sources of error. The RMS error of the velocity field is also computed, and velocity field error statistics can be computed.

3.8 Optimisation

Simulating PIV is computationally intensive, and considerable thought has been given to writing an efficient program. Python was chosen to program the digital twin as it has extensive support from scientific libraries, and is easy to use. A downside is that Python is an interpreted language meaning code is interpreted at runtime, which is inefficient. To circumvent this problem, the Numba library was used. Numba provides the function decorator '@njit' which tells the interpreter to compile the function into C the first time it is called. This causes the first function call to be very slow, but subsequent calls to be vastly sped up. Numba also provides function decorators which for running functions in parallel and with faster mathematical operations which sacrifice precision but at a resolution orders of magnitude higher than that required for accurate PIV simulation.

Particle marching is parallelised and compiled into C using Numba. This decreases the particle marching computation time by between one and two orders of magnitude, depending on the workload. Creation of the initial image, in which each particle position adds a single unit of brightness to the image, is performed using vectorised numpy operations, taking advantage of numpy's compiled and optimised functions running in C. The blurring of this image to create the Gaussian brightness distributions specified in the model is performed using 'scipy.gaussian_filter' which convolves a Gaussian kernel with the image. This function leads to image generation which is at least two orders of magnitude faster than performing the blurring when each pixel is added to the image, and it is for this reason that varying particle image diameters cannot be accommodated within the current model.

Spatial image cross correlation is the most computationally intensive part of the digital twin and dominates the runtime for a typical workload. Spatial image cross correlation by direct computation (iterating through all possible shift vectors) has a time complexity of $\mathcal{O}(n^4)$, so increasing resolution drastically increases runtime. This method was originally implemented due to its simplicity and the authors lack of knowledge of other cross correlation algorithms. The functions involved in spatial image cross correlation were compiled, run using 'fast-math' and in 'no-python' mode (i.e. functions compiled to C), and where appropriate were parallelised. Parallelisation and compilation of the direct computation

method lead to a reduction in runtime of just over two orders of magnitude for a typical workload. However, this method was still excessively slow and hindered progress in data generation and analysis.

Spatial image cross correlation can equivalently be performed by convolving a kernel (window 1) with an image (window 2) over the bounds of interest (the maximum displacement vector limits). A standard implementation of this method still has a time complexity of $\mathcal{O}(n^4)$. If convolution is instead computed by taking the fast Fourier transform (FFT) of the kernel and window and then multiplying the appropriate elements (convolution in the spatial domain is multiplication in the Fourier domain) then a spatial image cross correlation method with a time complexity of $\mathcal{O}(n^2 \log_2(n))$ can be achieved. Implementing this in the digital twin using SciPy’s ‘fftconvolve’ function yields an order of magnitude runtime reduction for a typical workload.

4 Hardware Review

4.1 Lasers

Figure (5) is a scatter plot of maximum instantaneous luminous power against price for a variety of illumination sources. The maximum instantaneous power of an illumination source must be sufficient to create particle images during the interframing time. The scattering of light by particles is highly dependent on particle diameter, but for a given PIV system the particle size is limited by the requirement for particles to accurately follow the flow.

Conventional PIV systems use pulsed lasers generating 532 nm light at very high intensities for a very short pulse. They typically deliver between 5 and 500 mJ of laser energy per pulse. Figure (5) shows that these lasers are expensive but have far superior maximum instantaneous power ratings than other illumination sources.

PIV cost could be reduced by using alternative illumination sources. Continuous lasers over an order of magnitude cheaper exist, but the maximum instantaneous power available is at least four orders of magnitude lower than the pulsed lasers. To compensate for this loss of instantaneous power the exposure time and particle diameter could be increased. Whether these system changes are viable depends on the PIV system of interest. For example, PIV in water may use large particles with density matched closely to the density of water. Matching the particle density to the fluid density leads to much smaller particle slip errors and hence PIV in water at low speeds is possible with continuous laser sources, exposures on the order of tens of microseconds, and large particles. The effects of changing hardware parameters are investigated in section 5.

PIV cost could also be reduced by using LEDs as an illumination source. LEDs with costs an order of magnitude lower than continuous lasers are available with luminous

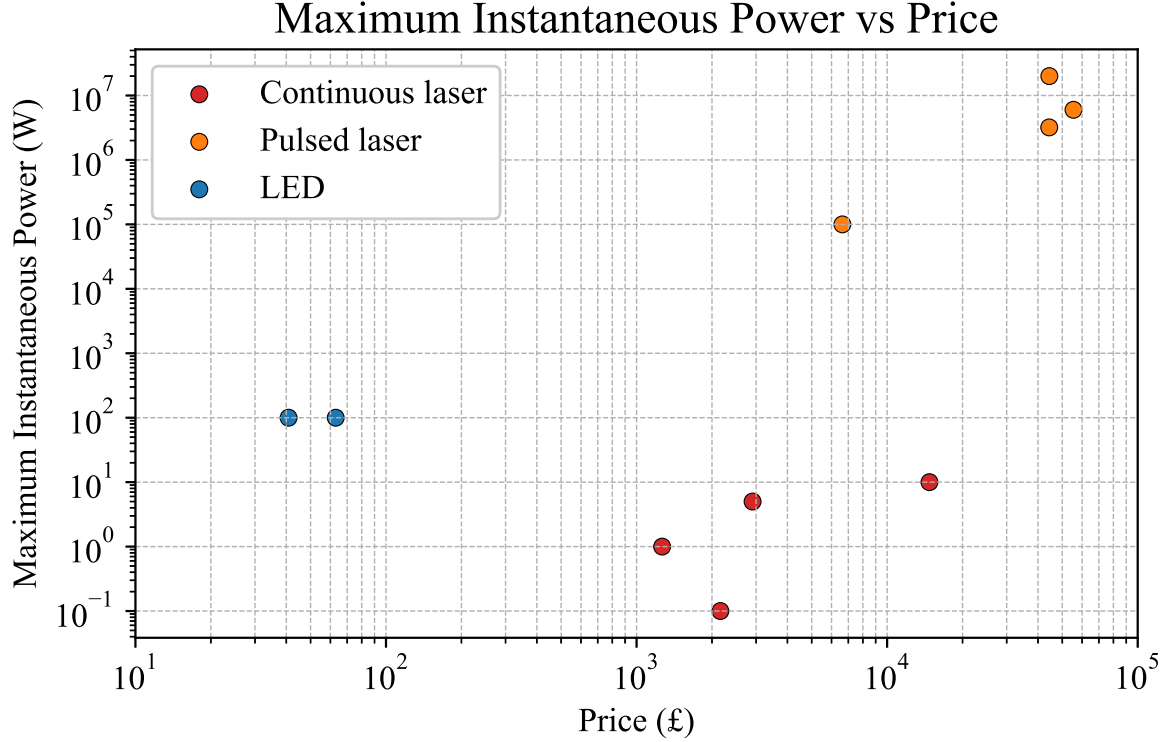


Figure 5: A scatter plot of maximum instantaneous luminous power against price for a variety of illumination sources.

powers an order of magnitude greater than continuous lasers. An array of LEDs with suitable optics could be used for PIV illumination, providing an excellent power to price ratio. However, the maximum instantaneous power available from an array of LEDs is limited by power supply constraints, cooling, and safety requirements.

4.2 Cameras

Figure (6) is a bubble chart showing camera resolution against interframing time, with the ‘bubbles’ scaled to have areas proportional to camera price. Conventional PIV systems use double shutter CCD cameras to achieve microsecond interframing times. A single CCD camera is included in figure (6) and represents the cheapest CCD camera found in the hardware review. Quotes were not received for other CCD cameras of interest, but it is known that they cost more than the plotted camera.

The issue with the CCD camera is cost. The plotted CCD camera was quoted at £7800, and alternative CCD cameras from the same supplier are known to be more expensive. An alternative camera option is an off the shelf CMOS camera. However, these cameras cannot double shutter image captures, in which two images are captured in quick succession, and instead their interframing time is dictated by their maximum frame rate. Therefore the interframing times achievable are at least three orders of magnitude

longer.

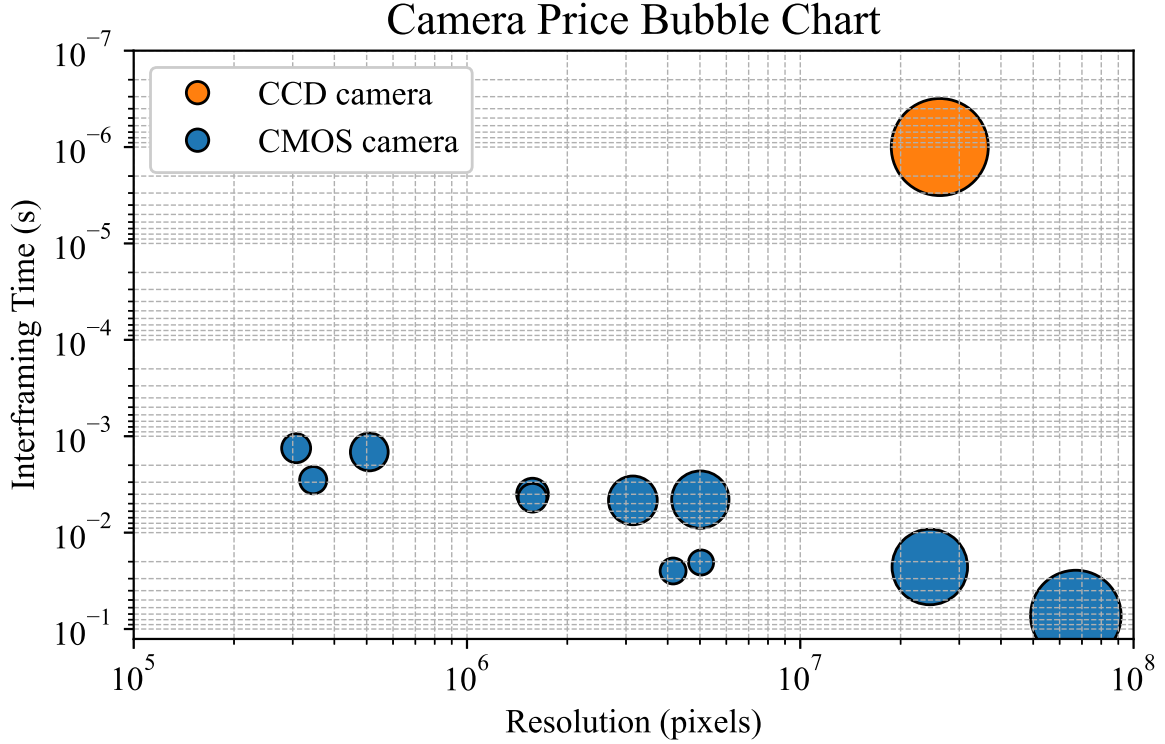


Figure 6: A bubble plot of camera resolution against interframing time, with bubble area proportional to camera price.

PIV using conventional CMOS cameras has been shown to be possible, but efforts have been limited to flows in water [4][10][11][14] or very slow gaseous flows.[5] Further discussion of the required hardware specifications for PIV is discussed in section (5).

5 Dimensional Analysis and Sensitivity Investigation

5.1 Introduction

The digital twin enables the analysis of PIV errors for a given PIV system. Characterising the sensitivity of PIV to variations in hardware parameters provides the information necessary for intelligent low cost PIV system design. In this section of the report, dimensional analysis is performed to determine the non-dimensional parameters which PIV RMS error magnitude depends on, and subsequently a sensitivity analysis will be performed by varying each non-dimensional variable about their value in a ‘typical’ PIV system.

PIV measurement errors are intrinsically random as the initial position of every particle affects the measurement error. For example, if two groups of particles happen to look similar and are close to each other a false correlation error is likely to occur. Hence to get

useful data the RMS measurement errors of a particular PIV system must be averaged over an ensemble of simulated PIV captures.

5.2 Dimensional Analysis and Typical PIV System Properties

Table (1) shows all of the digital twin input parameters for a full PIV simulation. The RMS error divided by the freestream velocity E_{RMS}/U is a dimensionless measure of average PIV measurement error. E_{RMS}/U must be a function of the properties in table (1) and the initial positions of every particle in the simulation. The initial particle positions are random and hence E_{RMS}/U is also random, but averages of E_{RMS}/U over a large number of samples are functions of the properties in table (1) only.

Particle Properties	Particle diameter	d_p	m
	Particle density	ρ_p	kg m^{-3}
Potential Flow Properties	Freestream velocity	U	m s^{-1}
	Cylinder diameter	d_c	m
	Flow dynamic viscosity	μ	$\text{kg m}^{-1} \text{s}^{-1}$
	Flow density	ρ_f	kg m^{-3}
Camera Properties	Resolution	R	pixels ²
	FOV centre x coordinate	x_c	m
	FOV centre y coordinate	y_c	m
	FOV size	w_0	m
Exposure Properties	Exposure time	t_e	s
	Interframing time	t_δ	s
	Illumination intensity factor	I	pixels ² s ⁻¹
	Particle image sigma	σ	pixels
Interrogation Properties	Maximum measurable velocity	U_{max}	m s^{-1}
	Spot size	w_i	m
	Spot overlap factor	N_o	-
Miscellaneous	Number of particles	n_p	-
	Gravity	g	kg m s^{-2}

Table 1: Digital Twin Input Parameters

Table (1) has nineteen parameters, the units have three dimensions (length, time, and mass) and hence Buckingham's Pi theorem states that sixteen independent non-dimensional groups can be formed. To simplify the analysis a few assumption are made.

Firstly, gravity is neglected. This is reasonable for flows where the gravitational forces on the particles are much smaller than the drag forces. Drag forces scale with d_p^2 and gravitational forces scale with d_p^3 , so gravitational forces are small compared to drag forces for the small (typically micrometer sized) particles involved in PIV measurements.

Secondly, we are interested in the sensitivity of PIV measurement errors to hardware parameters, not the sensitivity of PIV measurement errors to the position of the field of view in the specific flow used in the digital twin simulation. Hence the field of view centre

coordinates x_c and y_c can both be set equal to $d_c/2$ to centre the field of view on an interesting part of the flow travelling at the freestream velocity. This choice eliminates x_c and y_c as free variables, simplifying the analysis.

Finally, the maximum measurable velocity $U_{\max}$ (a property of the interrogation function) can be chosen sensibly in practice, and results from a poor choice of $U_{\max}$ are uninteresting. Hence $U_{\max}$ is set equal to $2.2U$ for the following sensitivity analysis. The maximum velocity involved in potential flow around a cylinder is $2U$ and hence $U_{\max} = 2.2U$ is a sensible choice to ensure the entire velocity range is captured and some headroom is provided for measurement errors that would occur in practice.

Setting $g = 0$, $x_c = y_c = d_c/2$ and $U_{\max} = 2.2U$ leaves fifteen independent dimensional input parameters. Therefore e_{RMS}/U depends on twelve non-dimensional parameters. Table (2) shows dimensionless groups chosen for the sensitivity analysis in this report. Many other choices of non-dimensional parameters exist, but these parameters were chosen to represent intuitive quantities with the hope of providing intuitive results.

The following sensitivity analysis was performed around a point defined by a typical PIV system. This typical PIV system has the specification given by table (3), with corresponding dimensionless parameters given by table (2). This typical system is a 2048×2048 pixel camera (4 megapixels) measuring flow at 30 m/s seeded with sub-micrometer oil droplets. The field of view is $75 \times 75 \text{ mm}^2$. The typical system camera, particle, and exposure time properties are taken from Adrian and Westerweel[2], and the remaining properties were estimated to produce images which look like real PIV images and to keep errors small. This system produces a median normalised RMS error of 3.7%.

Symbol	Definition	Description	‘Typical System’ Value
α	$\sqrt{R}/w_0$	Pixels density	2.731×10^4
Stk	$\frac{\rho_p d_p^2 U}{18 \mu d_c}$	Stokes number	1.036×10^{-4}
γ	ρ_p / ρ_f	Particle-Fluid density ratio	7.510×10^2
Re_c	$\frac{\rho_f U d_c}{\mu}$	Reynolds number for flow around a cylinder	1.430×10^5
ε	d_c / w_0	Ratio of cylinder diameter to field of view width	9.600×10^{-1}
N_I	$n_p \frac{w_i^2}{w_0^2}$	Image density (a dimensionless measure of particles per interrogation window)	1.200×10^1
N_o	N_o	Interrogation spot overlap factor	1.000
κ	w_i / w_0	Ratio of interrogation spot size to field of view size	3.125×10^{-2}
λ	$t_\delta U / d_c$	Ratio of distance travelled by the freestream during the interframing time to the cylinder diameter	1.667×10^{-2}
ξ	t_e / t_δ	Exposure to interframing time ratio	2.500×10^{-4}
ϕ	$\frac{\sigma_i w_0}{\sqrt{R} w_i}$	Ratio of particle image ‘radius’ to interrogation spot width	1.563×10^{-2}
ψ	$\frac{I t_e}{R}$	Characteristic image brightness from a particle	2.384×10^{-3}

Table 2: Non-dimensional input parameters for PIV sensitivity analysis

d_p	$3 \times 10^{-7} \text{ m}$
ρ_p	920 kg m^{-3}
U	30 m s^{-1}
d_c	$7.2 \times 10^{-2} \text{ m}$
μ	$1.85 \times 10^{-5} \text{ kg m}^{-1} \text{ s}^{-1}$
ρ_f	1.225 kg m^{-3}
R	4194304 pixels^2
w_0	$7.5 \times 10^{-2} \text{ m}$
t_e	$1 \times 10^{-8} \text{ s}$
t_δ	$4 \times 10^{-5} \text{ s}$
I	$1 \times 10^{12} \text{ pixels}^2 \text{ s}^{-1}$
σ_i	1 pixel
w_i	$2.34375 \times 10^{-3} \text{ m}$
N_o	1
n_p	12288

Table 3: Typical PIV System Dimensional Parameters

5.3 Sensitivity Analysis

5.3.1 Sensitivity to Pixel Density

Figure (7) is a 2D histogram showing the variation of the dimensionless PIV measurement error E_{RMS}/U with variations in the pixel density $\sqrt{R}/w_0$. Pixel density is defined as the width of the image in pixels divided by the width of the field of view in metres. It is a dimensionless measure of camera resolution. (A pixel is defined as the width of one of the squares commonly known as pixels in an image. Hence a pixel is a unit of length. A camera ‘pixel’ has an area of 1 pixel².)

The data for figure (7) is generated by sweeping through 40 values of pixel density equally spaced in a geometric series. The other dimensionless variables are held constant at their values for the typical PIV system of table (3). This approach of varying a single dimensionless group while holding all other dimensionless groups is performed throughout the sensitivity analysis. Figure (7) is generated by counting the number of generated data-points (with each data-point containing a value for the dimensionless parameter being investigated and a value for the normalised RMS error for that simulation) in each ‘bin’, where a ‘bin’ is a ‘rectangular region’ of the histogram plot. A colour is then assigned to this region based on the number of particles within the region, such that a rectangle with a light colour contains few simulated PIV results, and a rectangle with a dark colour contains many simulated PIV results. The black line in figure (7) displays the median normalised RMS error. The median is used as the sampling randomness intrinsic in simulated PIV (due to random particle initial positions) causes excessive noise in the mean data. Finally, the red cross represents the typical system and its median normalised RMS error.

The digital twin simulates PIV systems using the input parameters given in table (1), all of which are in SI units. The required dimensional input values are determined using the values of the dimensionless inputs, the relationships between the dimensionless inputs and the dimensional inputs, and the values of the field of view width, interframing time and dynamic viscosity. The three dimensional inputs are required as twelve dimensionless variables are derived from fifteen independent dimensional variables and therefore the dimensionless variables do not uniquely specify a PIV system. The dimensionless variables do of course dictate the performance of the PIV system, as expected from the dimensional analysis. This fact has been verified by varying the field of view width, interframing time and dynamic viscosity while holding the dimensionless variables constant.

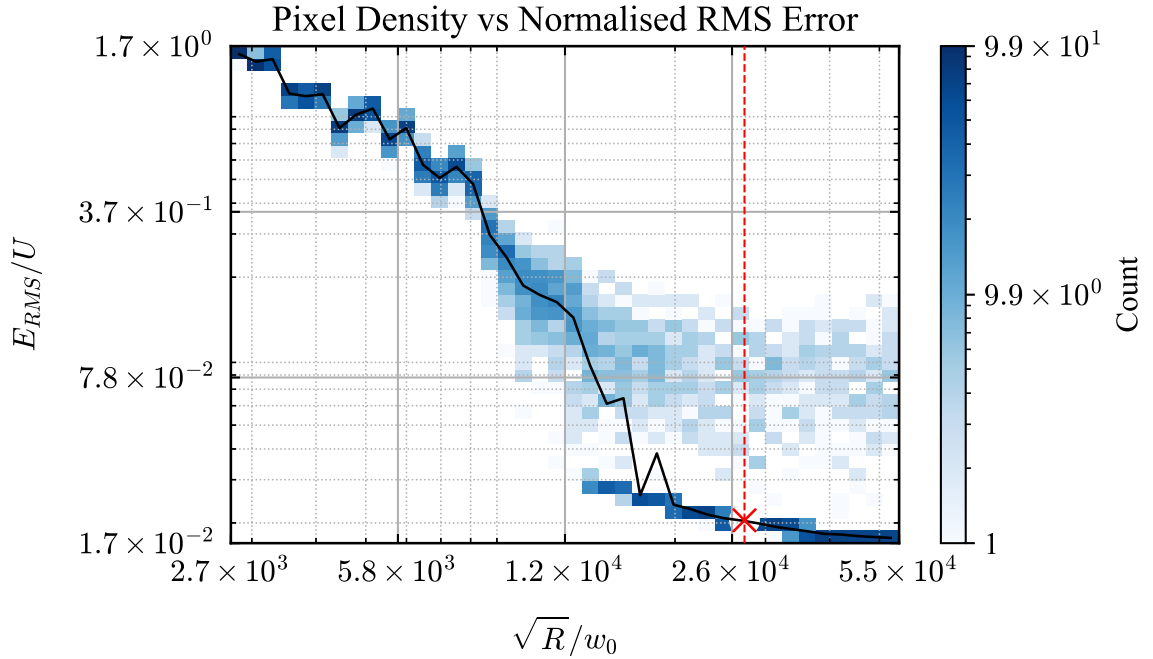


Figure 7: A 2D histogram and median trend line for normalised RMS error against pixel density for 4000 simulated PIV measurements at 40 values of pixel density.

At values of pixel density below 8.3×10^3 most of the individual particles are indistinguishable from each other, as seen in figures (8a) and (8b). Note that all images in figure (8) have an equal number of pixels and are sections of images with the same field of view. Therefore as $\sqrt{R}/w_0$ the images shown become proportionally smaller subsections of the full simulated image, effectively zooming in the shown view.

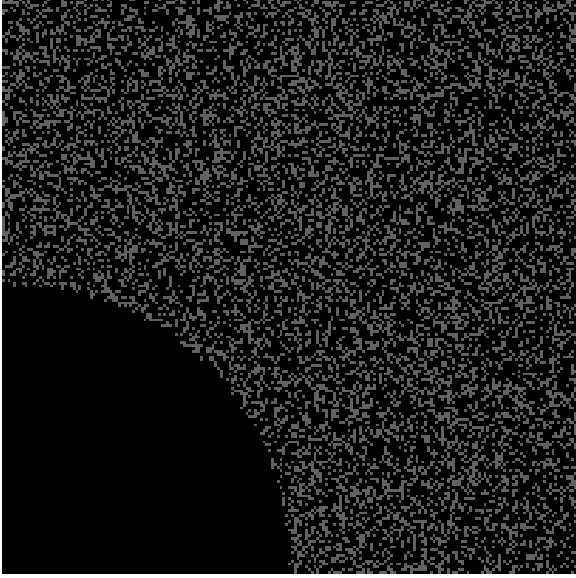
The particles being difficult to distinguish results in many false correlation errors, where the correlation peak is greatest for windows which contain different ‘similar looking’ groups of particles and hence the predicted displacement vector is not related to the displacement of the particles initially in the first window. These false correlation errors dominate the RMS error for small pixel densities.

Figure (9a) shows a simulated velocity field measurement for a pixel density of 2.7×10^3 , equal to the pixel density used for figure (8a). It is seen that most of the measured velocity vectors are large and point in random directions due to the quantity of false correlation errors. Those measurements which exhibit ‘true correlations’ between the two images have limited resolution due to the small amount of pixels travelled by the particles during the interframing time, and thus pixelisation errors (discussed in the following paragraph) are large. (For example, the velocity vectors in the region surrounding (0.0525 m, 0.042 m) are all identical due to pixel locking.)

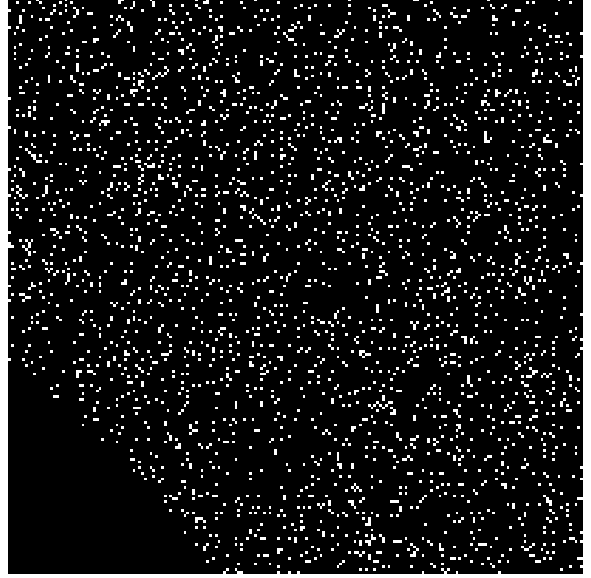
Increasing pixel density above 8.3×10^3 results in individually identifiable particle images, leading to a rapid decrease in the frequency of false correlation errors and the increase in the relative importance of pixelisation errors. Pixelisation describes the phenomena of continuous position values (the exact particle positions) mapping to a discrete set of position values (the ‘pixels’ in the camera). This causes a loss of information and means the position of a particle can only be determined to within plus or minus half a pixel.¹ Between pixel densities of 8.3×10^3 and 1.8×10^4 both false correlation errors and pixelisation errors are important. Figure (8c) shows an image in this regime in which most particles are individually distinguishable but are highly pixelated.

Above 1.8×10^4 the median simulated PIV error is dominated by the pixelisation error, which continues to decrease as pixel density increases. Eventually the reduction in error will plateau due to the presence of other sources of error. Figure (8d) shows a section of an image with a high pixel density. The particle images are clearly distinguishable and span multiple pixels, satisfying the design rule that “there should be at least 3-5 pixels per particle image diameter”.^[9] Figure (9b) shows a simulated velocity measurement with a pixel density of 5.5×10^4 . The measured velocity field looks as expected for potential flow around a cylinder, and the largest source of RMS error is particle slip. This is inferred from the fact that the error magnitudes are larger in regions of greater velocity. (It is also confirmed by direct simulation of PIV with a pixel density of 5.5×10^4 and particles with a lower Stokes number.) A single false correlation error is present at approximately (0.015 m, 0.033m).

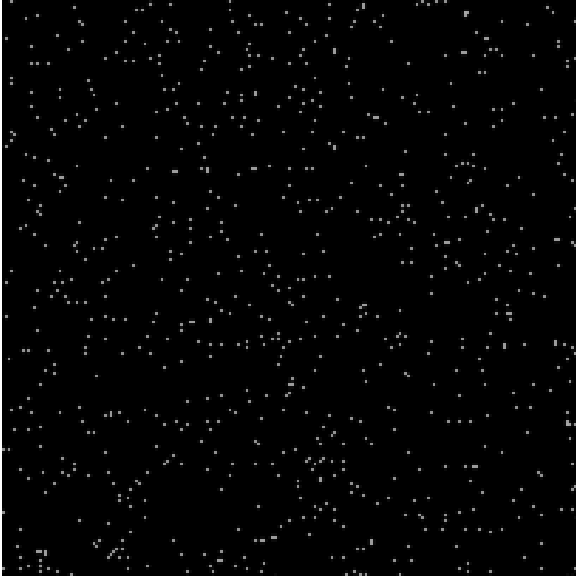
¹Strictly the statement that particle position can only be determined to within plus or minus half a pixel is not true as the difference in brightness between pixels which are equally spaced from the ‘central pixel’ in a particle image encodes information. Interrogation algorithms have been developed with sub-pixel accuracy, but these are not implemented in the digital twin due to constraints on computational power and time. See Lecordier et al.^[6] for an example of a sub-pixel interrogation algorithm.



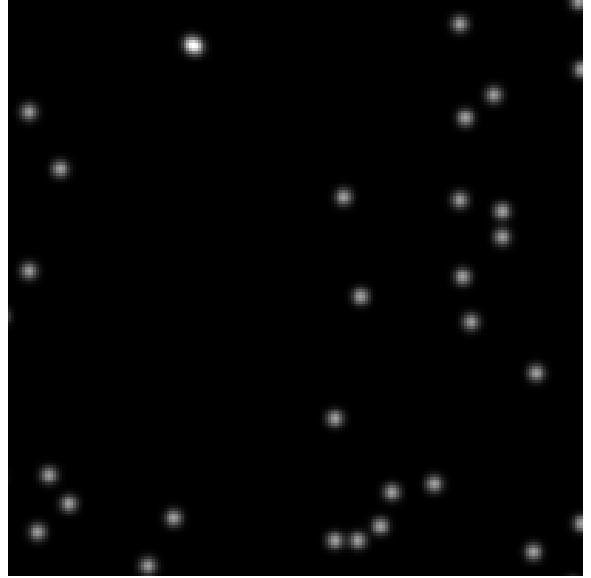
(a) $\sqrt{R}/w_0 = 2.7 \times 10^3$, artificially brightened



(b) $\sqrt{R}/w_0 = 5.8 \times 10^3$, artificially brightened

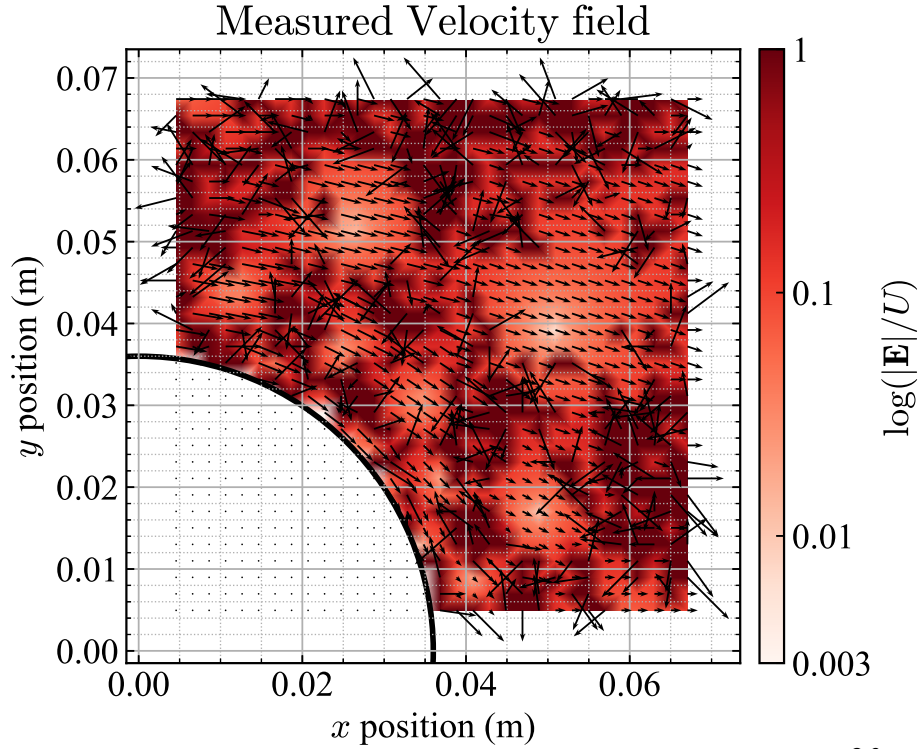


(c) $\sqrt{R}/w_0 = 1.2 \times 10^4$

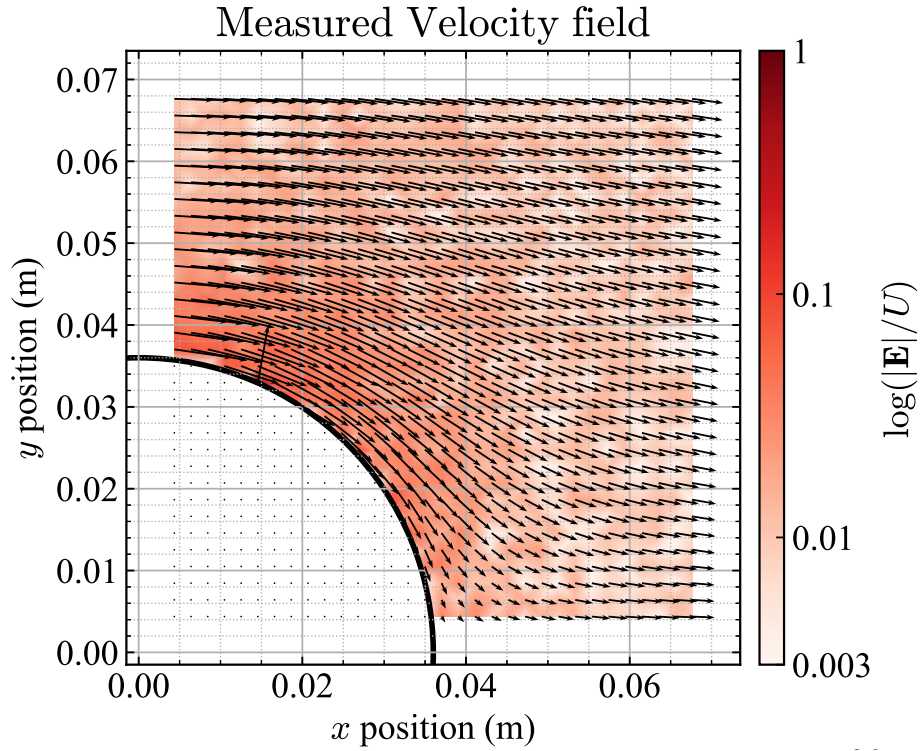


(d) $\sqrt{R}/w_0 = 5.5 \times 10^4$

Figure 8: Synthetic images generated by the digital twin for four different values of pixel density, with the other non-dimensional parameters held constant. Figures (8a) and (8b) have artificially increased brightnesses to enable a human to see them properly. Figure (8a) is a complete simulated image, whereas figures (8b), (8c) and (8d) are cropped sections of larger images with the same field of view as figure (8a), centred on the same centrepoint as (8a), so that figures (8a), (8b), (8c) and (8d) all have an equal number of pixels. (I.e. The images from which each figure is taken have the same field of view, but as $\sqrt{R}/w_0$ increases the fraction of the image shown in the report is reduced.)



(a) A simulated velocity field measurement for pixel density equal to 2.7×10^3 , coloured to show the normalised magnitudes of measurement errors.



(b) A simulated velocity field measurement for pixel density equal to 5.5×10^4 , coloured to show the normalised magnitudes of measurement errors.

Figure 9: Simulated velocity field measurements with varying pixel density

5.3.2 Sensitivity to Stokes Number

Figure (10) is a 2D histogram showing the variation of dimensionless measurement error against Stokes number. Stokes number is the ratio of a characteristic time of a particle to a characteristic time of a flow. For simulated PIV measurements around a cylinder, Stokes number is the ratio of the characteristic relaxation time of a particle to the time taken to travel one cylinder diameter at the free-stream velocity, given mathematically by equation (8).

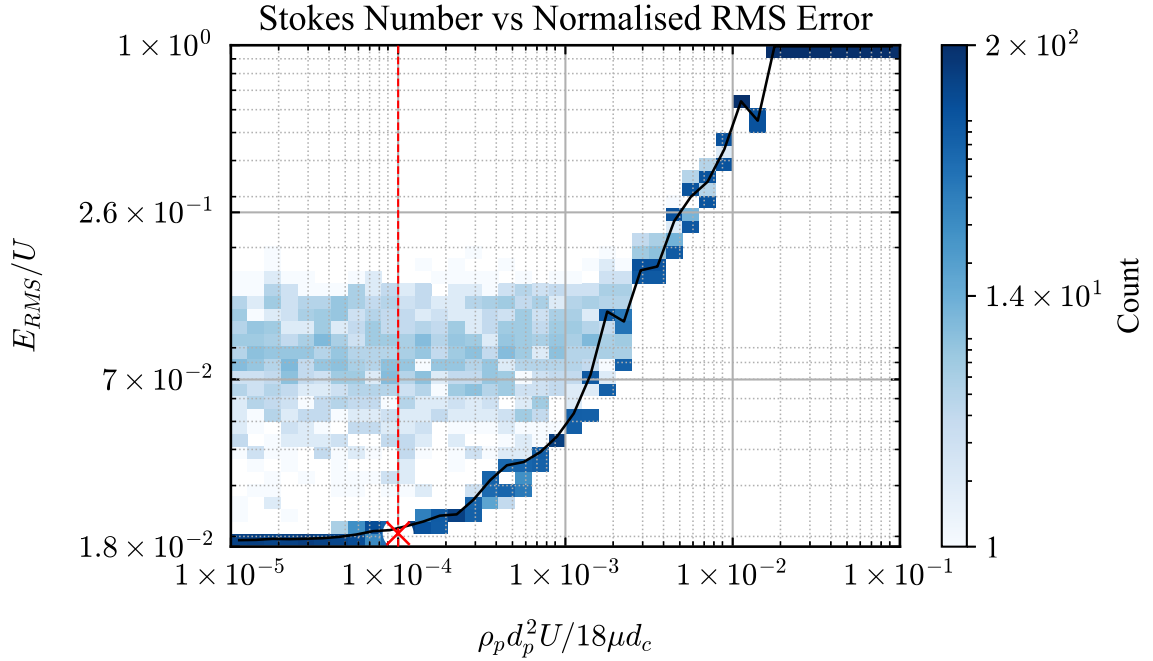


Figure 10: A 2D histogram and median trend line for normalised RMS error against Stokes number for 4000 simulated PIV measurements at 40 values of Stokes number.

$$\text{Stk} = \frac{\rho_p d_p^2 U}{18 \mu d_c} \quad (8)$$

Stokes number is a measure of tracer particle fidelity. For very low Stokes numbers the particles should follow the flow almost perfectly leading to miniscule particle slip errors, whereas for large Stokes numbers the particles will experience large slip velocities resulting in large measurement errors. Figure (10) shows the expected relationship. At low values of Stokes number ($\text{Stk} < 5 \times 10^{-5}$) the particles experience very little slip and other sources of error dominate. Between $\text{Stk} = 5 \times 10^{-5}$ and $\text{Stk} = 1.8 \times 10^{-2}$ the median normalised RMS measurement error increases due to increasing particle slip. Above $\text{Stk} = 1.8 \times 10^{-2}$ the normalised measurement error is constant because while the particles do not follow the flow accurately, they do have velocities associated with the flow, and therefore there is an upper bound of the RMS error. (e.g. A normalised error

of 2 corresponds to a particle in the free-stream travelling upstream at the free-stream velocity. This is clearly unlikely.)

5.3.3 Sensitivity to Density Ratio

Figure (11) is a 2D histogram showing the variation of dimensionless measurement error against the ratio of particle density to fluid density. The model used for this sensitivity analysis has zero gravity and therefore weight effects are neglected. It is therefore unsurprising that figure (11) shows no variation in normalised RMS error large enough to attribute to anything but measurement noise.

If gravity was included in the model, weight would cause particle motion and therefore contribute to system errors. In practice, the largest accelerations experienced by particles during a PIV measurement tend to be significantly greater in magnitude than gravitational acceleration. Therefore errors due to weight are smaller than errors due to particle slip and hence a particle with appropriate Stokes number will yield good tracer fidelity regardless of the ratio of particle density to fluid density. I.e. The density ratio does not affect tracer fidelity provided the Stokes number is held constant.

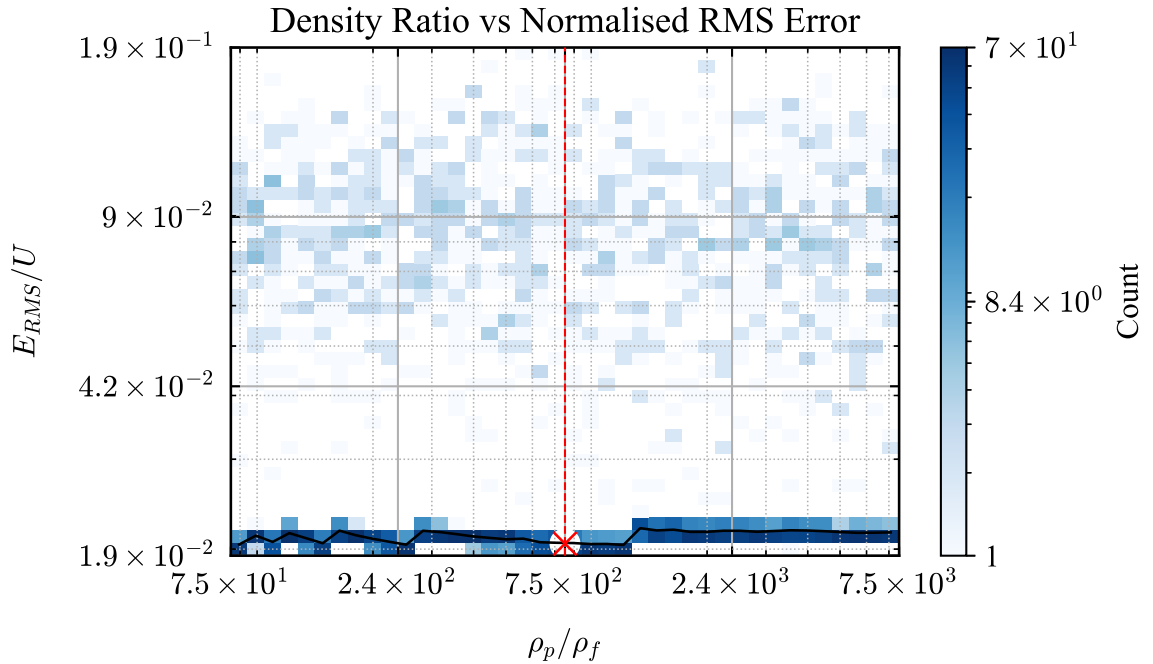


Figure 11: A 2D histogram and median trend line for normalised RMS error against particle to fluid density ratio for 4000 simulated PIV measurements at 40 values of particle to fluid density ratio.

5.3.4 Sensitivity to Reynolds Number

Figure (12) is a 2D histogram showing the variation of dimensionless measurement error against the Reynolds number for the flow around the cylinder. The digital twin uses potential flow as the underlying flow field, and hence varying Reynolds number does not change the shape of the underlying flow field. Therefore, changing the Reynolds number should not affect the normalised RMS error magnitude, provided the other dimensionless variables are held constant. This reasoning is validated by figure (12) showing that the normalised RMS error distribution is invariant (disregarding effects from finite random sampling) with Reynolds number.

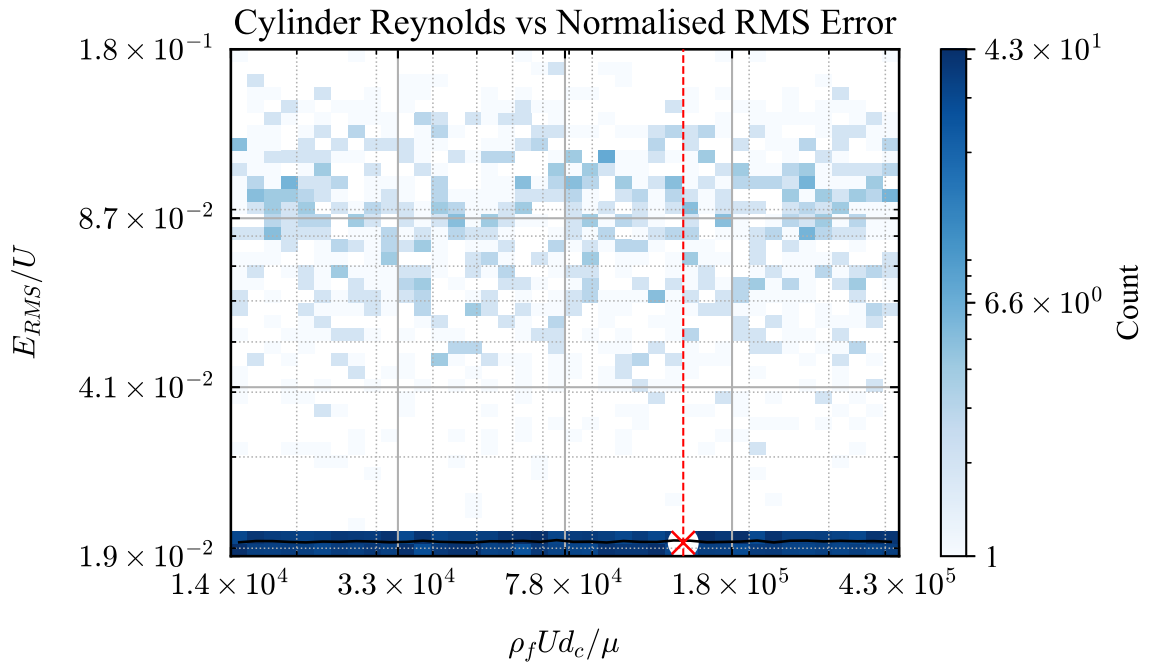


Figure 12: A 2D histogram and median trend line for normalised RMS error against the Reynolds number for flow around a cylinder, for 4000 simulated PIV measurements at 40 values of the Reynolds number for flow around a cylinder.

5.3.5 Sensitivity to Cylinder Diameter to FOV Width Ratio

Figure (13) is a 2D histogram showing the variation of dimensionless measurement error against the ratio of cylinder diameter to field of view width. The median error is small for values of d_c/w_0 between 1 and 1.25. Decreasing d_c/w_0 below 1 leads to a steadily increasing median normalised RMS error magnitude, and increasing d_c/w_0 above 1.25 leads to a sharp increase in median normalised RMS error magnitude.

As d_c/w_0 becomes small, the distance travelled by particles during the interframing time also becomes small because $U t_\delta / d_c$ (the ratio of the distance travelled by a particle

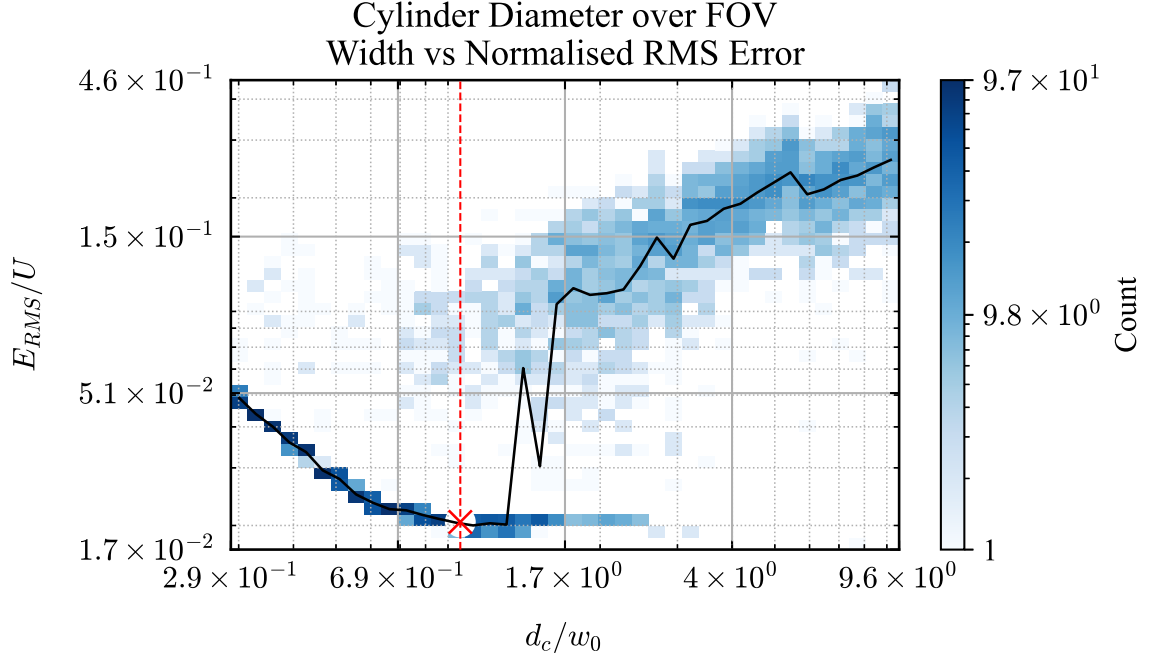
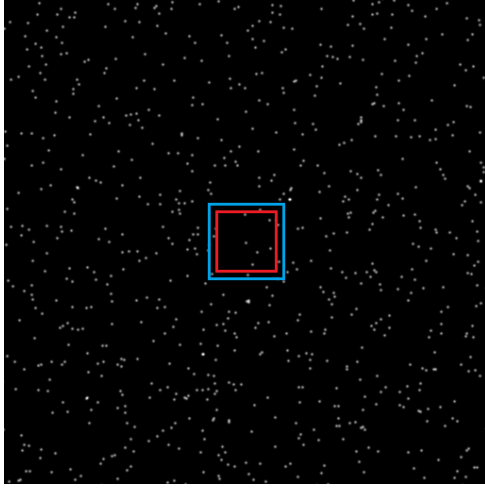


Figure 13: A 2D histogram and median trend line for normalised RMS error against the ratio of cylinder diameter to field of view width, for 4000 simulated PIV measurements at 40 values of the ratio of cylinder diameter to field of view width.

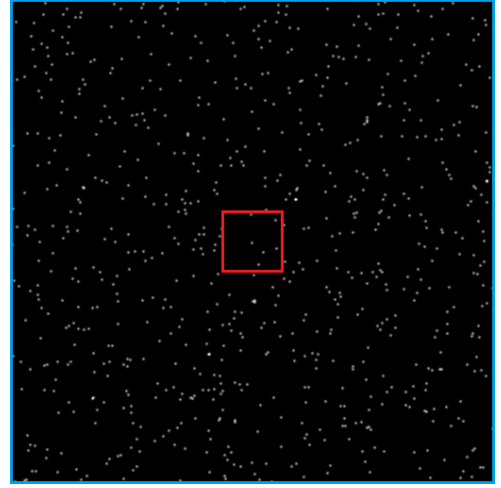
in the freestream during the interframing time to the diameter of the cylinder) is held constant. Therefore the distance travelled by a particle at the maximum measurable velocity (set to 2.2 times the freestream velocity) during the interframing time is small, and hence the maximum shift vector (see section 3.6 for the definition of a shift vector) used to search for matching particles in the second PIV image is a small number of pixels. This is illustrated in figure (14a), where the red box represents the size of an interrogation spot, and the blue box represents the extents searched by the spatial image cross correlation algorithm. Note that the searched extents in the second image will always include the particles which lie inside the interrogation spot in the first image, and that the blue box being small indicates the particles travelling only a short distance. The small maximum shift vector results in highly quantised velocity measurements due to the small number of candidate shift vectors and the inability of this spatial image cross correlation algorithm to determine the appropriate shift vector with sub-pixel accuracy.

When d_c/w_0 becomes large the distance travelled by particles during the interframing time also becomes large, requiring a large region to be searched in the second image for the particles in the interrogation spot in the first image. This increases the change of false correlation errors, leading to the increase in median normalised RMS error seen for cylinder diameter to field of view width ratios greater than 1.25.

Finally, note that the dimensionless parameter d_c/w_0 sets the size of the flow feature



(a) $d_c/w_0 = 0.1$



(b) $d_c/w_0 = 3$

Figure 14: Two equally sized sections of synthetic PIV images with the interrogation window size marked in red and the size of the region searched for a matching interrogation window in the second PIV image marked in blue. The images have equal fields of view, numbers of pixels, and interrogation spot sizes, but different cylinder sizes (outside of the section shows) and maximum interrogation spot shift vectors.

(the cylinder) relative to the field of view size, while another dimensionless parameter w_i/w_0 sets the size of an interrogation window relative to the field of view size. Together, these parameters set the size of an interrogation window relative to the size of the flow feature. These three parameters have interlinked physics, but choosing any two fixes the third. This explains why figure (18) has some similar features to figure (13).

It is recommended to operate the ‘typical system’ with a cylinder diameter to field of view width ratio of 1 to minimise both pixelisation and false correlation errors. Increasing pixel density (i.e. using a camera with greater resolution) can pull the regions of increasing pixelisation error and increasing false correlation error further apart resulting in a more robust and versatile system.

5.3.6 Sensitivity to Image Density

Figure (15) is a 2D histogram showing the variation of dimensionless measurement error against the image density. The image density is the mean number of particles in each interrogation window. Figure (16) shows four 128×128 pixel² sections of simulated PIV images from systems with particle image densities of 1, 12, 400 and 1000, and other dimensionless parameters set to the ‘typical values’ given in table (2).

At low values of image density the median normalised RMS error is large because most interrogation spots do not contain enough particles to accurately determine the mean velocity of the interrogation spot. Figure (16a) shows a section of a PIV image for a system with an image density of 1. For this system, each interrogation spot contains an average of one particle image, and hence interrogation spot is unlikely to have unique

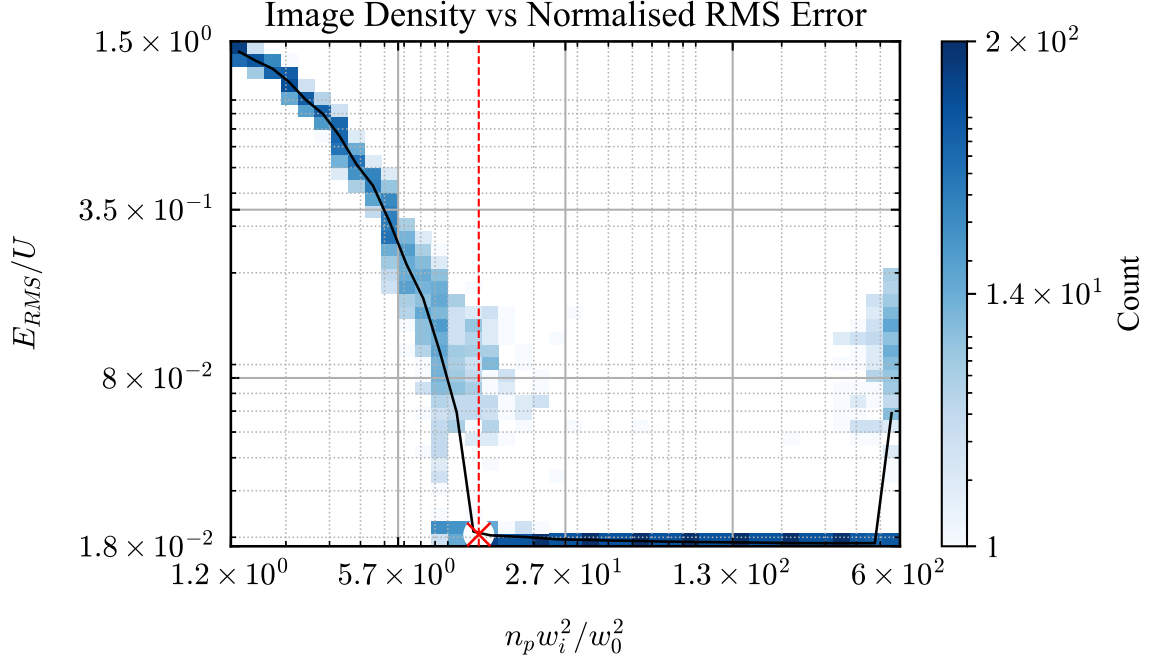


Figure 15: A 2D histogram and median trend line for normalised RMS error against image density for 5400 simulated PIV measurements at 54 values of image density.

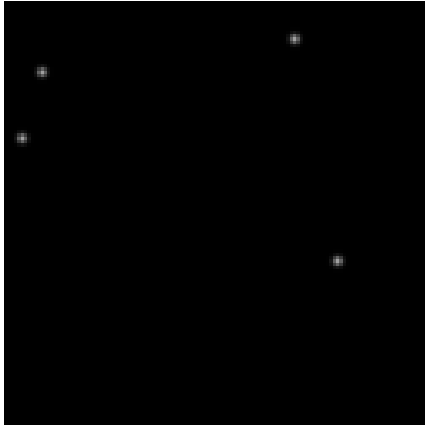
features which allow its location to be identified in the second image. An interrogation spot containing only one particle image will produce a correlation peak for any shift vector which maps the particle image onto any particle image in the second image. Hence the interrogation spot shift vector cannot be confidently determined unless there are no other particles near the original interrogation window.

In the case of image density much less than unity, individual particles can be tracked between images, but this requires a different approach to spatial image cross correlation. This is the basis of particle tracking velocimetry (PTV), but PIV is preferred to PTV due to the greater achievable velocity measurement density.

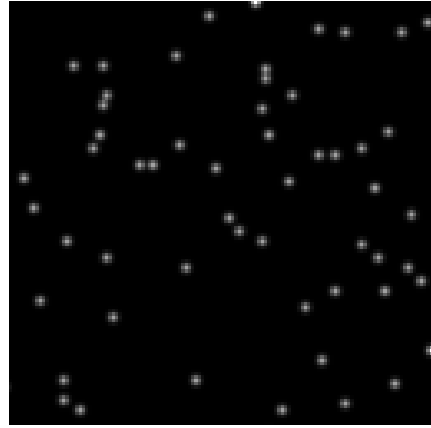
For values of image density between 12 and 450 the median image has no false correlation errors, and hence the normalised RMS error is low and is governed entirely by pixelisation and particle slip errors. Figures (16b) and (16c) show sections of simulated PIV images for systems with image densities of 12 and 400 respectively. In figure (16b) particles are individually identifiable, and enough particles exist to track ‘unique looking’ interrogation spots between images. In figure (16c) there are too many overlapping particle images to individually identify particles. Nevertheless, accurate spatial image cross correlation is possible due to the particle image speckle patterns being uniquely identifiable across images. This is the basis of particle speckle velocimetry (PSV), which is differentiated from conventional PIV only by the choice of high image densities. Between image densities of 23 and 310 the digital twin simulated zero false correlation errors. Therefore

this is the range of optimal particle image density for PIV using the otherwise ‘typical’ system defined in table (2). The ‘typical system’ parameters were decided before the sensitivity analysis and lie outside of this range. Moving the image density into this range is expected to lead to fewer false correlation errors in all sensitivity sweeps and a more robust system overall. Identifying possible parameter improvements, such as increasing image density, is a key application of the digital twin.

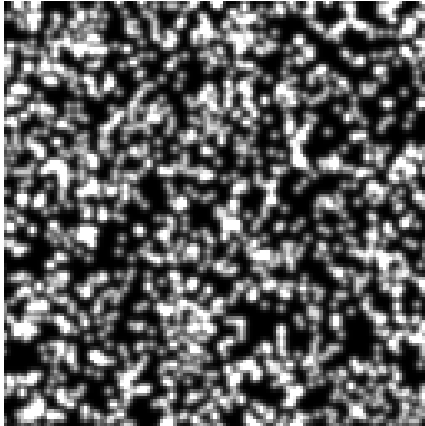
At values of image density greater than 400 the frequency of false correlation errors increases rapidly, and hence the normalised RMS measurement error also increases rapidly to an unacceptable magnitude. Figure (16d) shows a section of an image from a simulated PIV measurement with a ‘typical’ system (as defined by table (2) modified to have an image density of 1000. The normalised RMS error magnitude is large for the obvious reason that individual particles or particle speckle patterns are impossible to track between images.



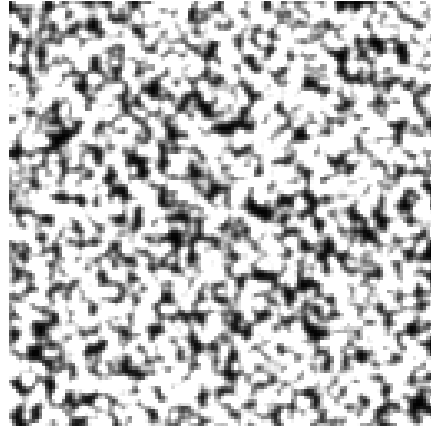
(a) Image Density $n_p w_i^2 / w_0^2 = 1$



(b) Image Density $n_p w_i^2 / w_0^2 = 12$



(c) Image Density $n_p w_i^2 / w_0^2 = 400$



(d) Image Density $n_p w_i^2 / w_0^2 = 1000$

Figure 16: 128×128 pixel² sections of simulated PIV images from systems with varying particle image densities and otherwise ‘typical’ dimensionless parameters as given in table (2). These image sections each contain four interrogation spots and hence each contain approximately $4(n_p w_i^2 / w_0^2)$ particles.

5.3.7 Sensitivity to Interrogation Spot Overlap

Figure (17) is a 2D histogram showing the variation of dimensionless measurement error against interrogation spot overlap factor. A spot overlap factor of 1 maximises spatial resolution for independent velocity measurements. Spot overlap factors greater than 1 yield interdependent velocity measurements, but spot overlap factors between 2 and 4 are often used in practice to increase spatial resolution. Spot overlap factors below 1 unnecessarily reduce spatial resolution and should not be used.

The variation in figure (17) is attributed entirely to random sampling statistics and does not reflect any change in the mechanisms causing PIV measurement errors. Low spot overlap factors lead to a small number of velocity vector measurements resulting in greater variance of the sampled normalised RMS error distribution. Large spot overlap factors lead to many velocity vector measurements, increasing the chance of a false correlation error occurring. These effects are both related to sampling statistics rather than PIV measurement error mechanisms.

Interrogation spot overlap factors between 1 and 4 inclusive should be used in practice to achieve good spatial resolution without having excessively interdependent velocity vector measurements.

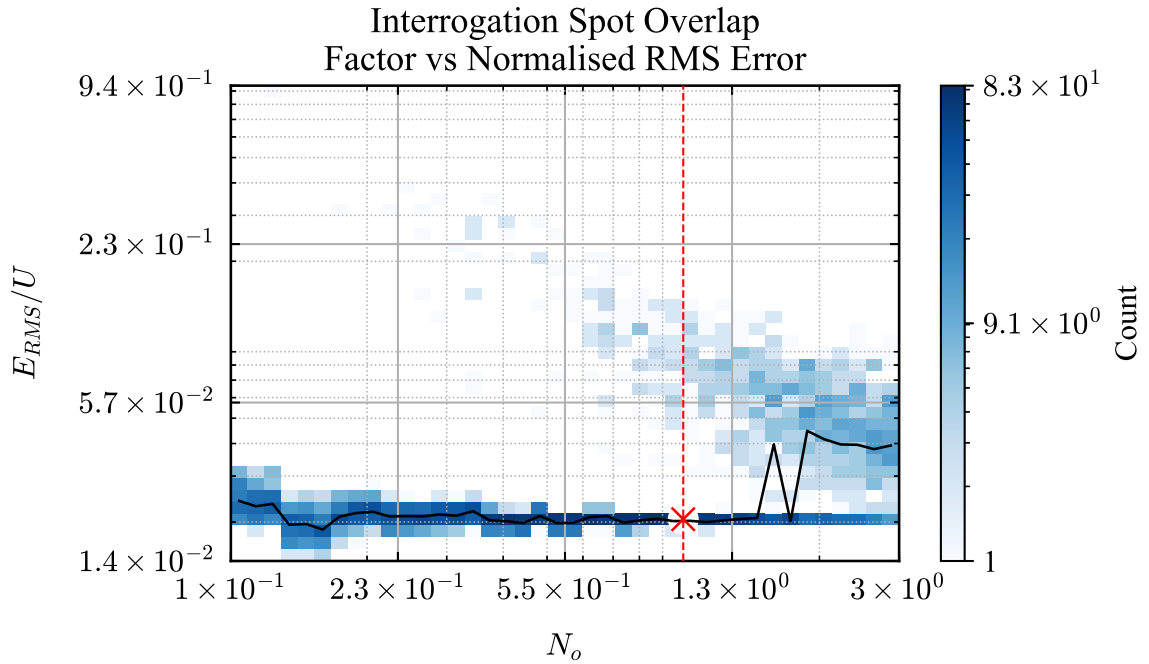


Figure 17: A 2D histogram and median trend line for normalised RMS error against the interrogation spot overlap factor for 4000 simulated PIV measurements at 40 values of the interrogation spot overlap factor.

5.3.8 Sensitivity to Interrogation Spot to FOV Width Ratio

Figure (18) is a 2D histogram showing the variation of dimensionless measurement error against the ratio of the interrogation spot width to the field of view width. An interrogation spot to field of view width ratio of $w_i/w_0 = 2.7 \times 10^{-2}$ gives the lowest median normalised RMS error, with median normalised RMS error increasing sharply for deviations below this value and increasing slowly for deviations above this value.

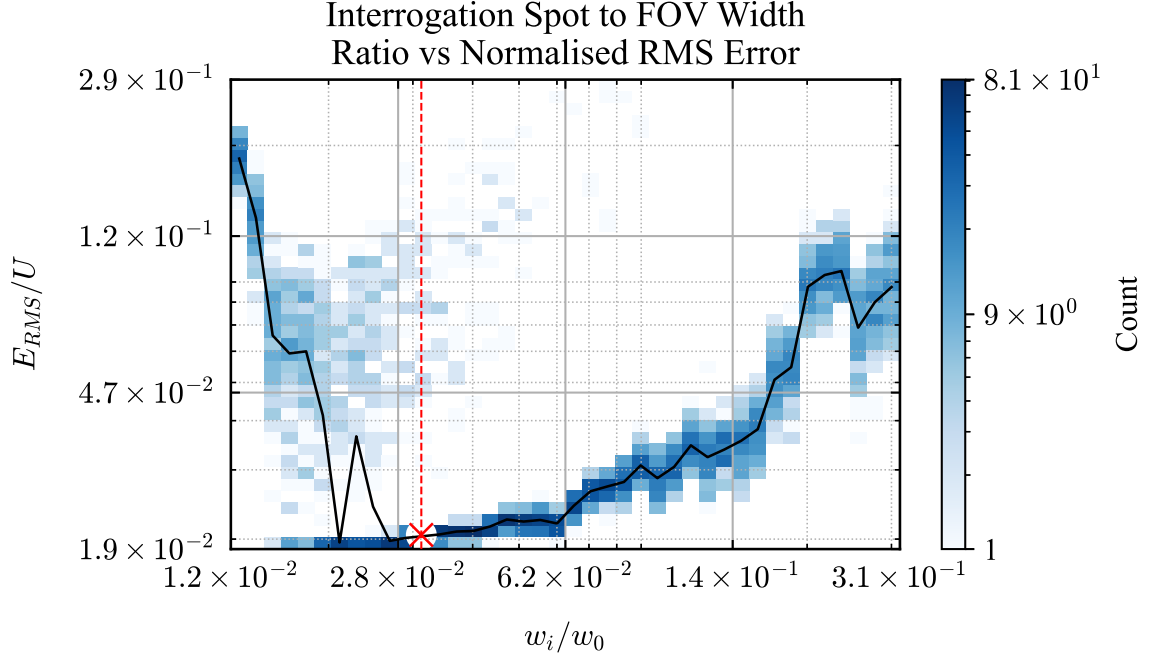


Figure 18: A 2D histogram and median trend line for normalised RMS error against the ratio of interrogation spot width to field of view width, for 4000 simulated PIV measurements at 40 values of the ratio of interrogation spot width to field of view width.

The ratio of particle image diameter (i.e. twice the standard deviation of the 2D Gaussian distribution representing the particle image) to interrogation spot width is held constant, so for small interrogation spot to field of view width ratios the particle images become small and collapse to single pixels. The interrogation window also becomes small relative to the field of view and relative to the extents searched for correlated particles as Ut_δ/d_c is held constant. Figure (19a) shows the extents searched for the interrogation window indicated in red for a synthetic PIV image from a system with $w_i/w_0 = 1.3 \times 10^{-2}$ and otherwise typical parameters. The searched image extents are much larger than the red interrogation window and the particle images are pixelated, leading to false correlation errors. The interrogation spot to FOV width ratio must be kept above 2.7×10^{-2} to avoid frequent false correlation errors for the otherwise ‘typical’ system.

Note that the image sections in figure (19) are different sizes. Importantly, the red interrogation window increases in size from figure (19a) to figure (19c), but the maximum

shift vector (see section 3.6 for the definition of a shift vector) used to search for correlated particles remains constant and therefore the number of pixels between the image section edge and the red interrogation spot also remains constant. The most important takeaway from figure (19) is that as w_i/w_0 becomes larger the interrogation spot size becomes larger relative to the extents searched for correlated particles.

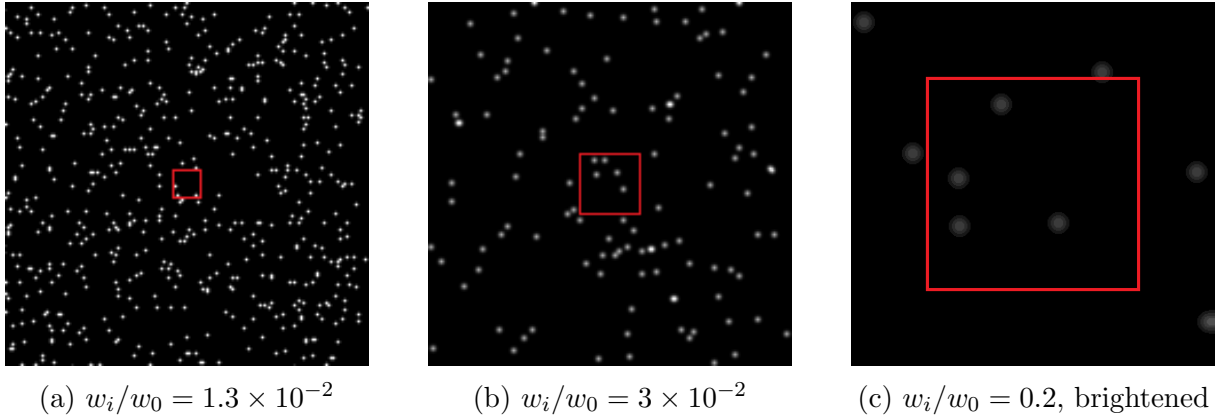


Figure 19: Sections of synthetic PIV images for different values of the ratio of interrogation spot width to field of view width. The red boxes indicate the size of an interrogation window, and the image size equals the size of the extents searched for particles correlated with the interrogation window.

For large interrogation spot to FOV width ratios the particles are clearly visible and span multiple pixels, as seen in figure (19c) which has been artificially brightened after simulation for ease of viewing. The interrogation spot has a size similar to the extents searched for correlated particles, so false correlation errors occur at negligible frequency. Quantisation errors (also known as pixelisation errors) are also small as the maximum shift vector magnitude is large [73 pixels in figure (19c)] and therefore the velocity measurement resolution is high. Despite this the median normalised RMS error increases with w_i/w_0 for values of w_i/w_0 greater than 2.7×10^{-2} . The errors for large interrogation spot to field of view width ratios are due to the mean velocity of particles within the large interrogation spots not being representative of the velocity at the centre of the interrogation spot. I.e. The velocity difference between two locations within the same interrogation spot is significant. Hence the velocity measured does not necessarily correspond to the velocity at the centre of the interrogation spot.

It is recommended to operate the otherwise ‘typical system’ at an interrogation spot to field of view width ratio of 3.5×10^{-2} , as this is large enough to keep the frequency of false correlation errors very low, but small enough to introduce only small errors due to velocity differences across the interrogation spot.

5.3.9 Sensitivity to Freestream Interframing Distance over Cylinder Diameter

Figure (20) is a 2D histogram showing the variation of dimensionless measurement error against the ratio of the distance travelled at the freestream velocity during the interframing time to the cylinder diameter. The ratio of cylinder diameter to field of view width is fixed by the dimensionless variable d_c/w_0 and the size of a pixel is fixed by the pixel density $\sqrt{R}/w_0$. Therefore decreasing the ratio of the distance travelled at the freestream velocity during the interframing time (also known as the freestream interframing distance) to the cylinder diameter decreases the number of pixels travelled during the interframing time by the particles. This leads to pixelisation errors for small values of Ut_δ/d_c due to the low resolution of the velocity measurement. Pixelisation errors are discussed extensively in section 5.3.1.

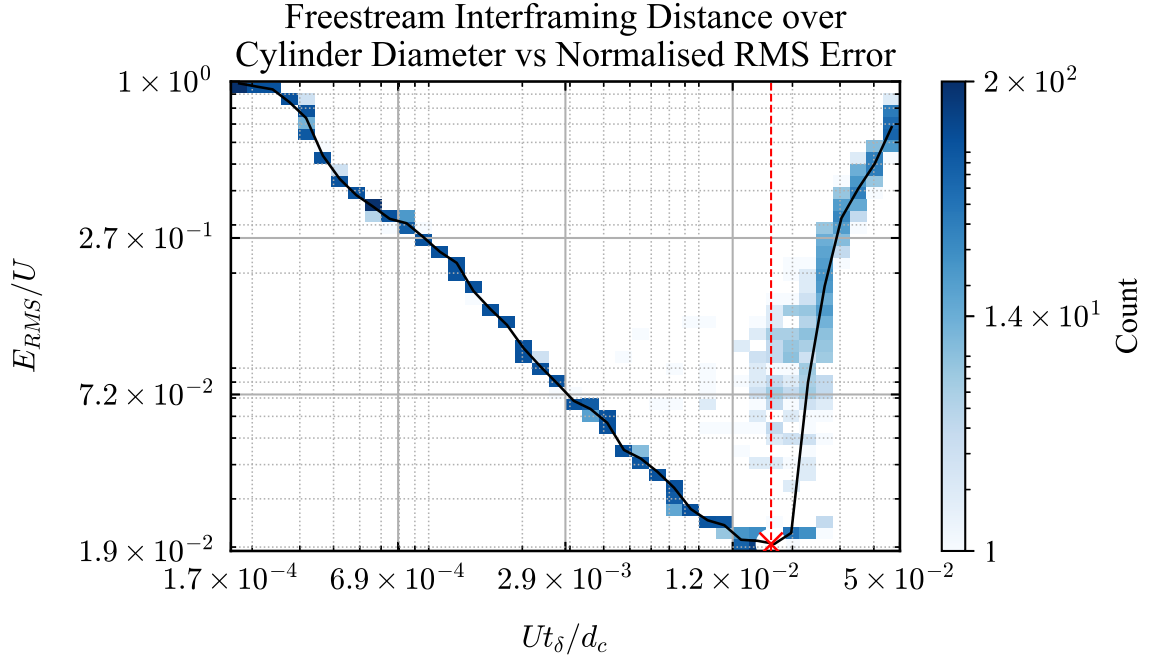


Figure 20: A 2D histogram and median trend line for normalised RMS error against the ratio of the distance travelled at the freestream velocity during the interframing time to the cylinder diameter, for 5000 simulated PIV measurements at 50 values of the aforementioned ratio.

For large ratios of the freestream interframing distance to the cylinder diameter the distance travelled by particles during the interframing time is large relative to the size of an interrogation spot. The large extents searched for matching particles make false correlation errors more likely, as described in section 5.3.8. Additionally, shear in the flow distorts the interrogation spot between images, increasing false correlation errors in regions of high shear.

Figure (21) shows two sections of consecutive simulated PIV images for a system with $w_i/w_0 = 3 \times 10^{-2}$ and otherwise ‘typical’ parameters. Figures (21a) and (21b) have identical fields of view with the cylinder sitting at the bottom left of both images. The red box indicates an interrogation spot tracking the same group of particles between images. The effect of flow shear on the particles can be seen by inspecting the particles within the interrogation spot in both images. The particle in the bottom left of the interrogation spot is identically positioned relative to the interrogation spot in both images, providing a useful reference point. The pair of particles in the top left of the interrogation spot in figure (21a) is seen to partially exit the interrogation spot in figure (21b), and the particle just outside the bottom right corner of the interrogation spot in figure (21a) is seen to enter the interrogation spot in figure (21b). Therefore the relative positions of the particles has changed in the interframing time. This change in relative positions causes the interrogation spots to look different to each other, lowering the correlation peak and increasing the likelihood of false correlation errors in regions of large shear. Regions of large rotation would produce a similar effect, but the digital twin uses irrotational potential flow so these effects are not seen. Improved correlation algorithms capable of sub-pixel accuracy and the measurement of shear and rotation exist [6], but their computational time complexity makes them unsuitable for use in the digital twin.

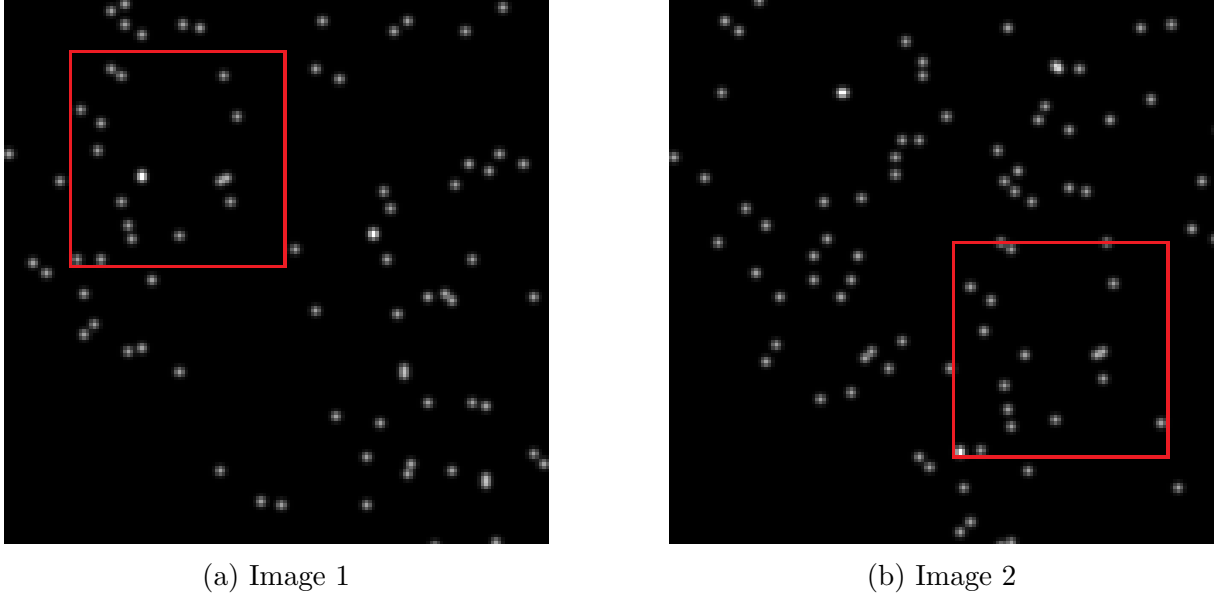


Figure 21: Identical field of view sections of the two consecutive simulated PIV images used to make a single velocity measurement for a system with $w_i/w_0 = 3 \times 10^{-2}$ and otherwise ‘typical’ parameters. A red box indicates an interrogation window tracking a group of particles near the cylinder between the two images.

Figure (22) shows a simulated velocity field measurement for a system with a freestream interframing distance to cylinder diameter ratio of 3×10^{-2} . As expected from the preceding discussion, the large freestream interframing distance to cylinder diameter ratio leads to frequent false correlation errors in the region of high shear near the cylinder.

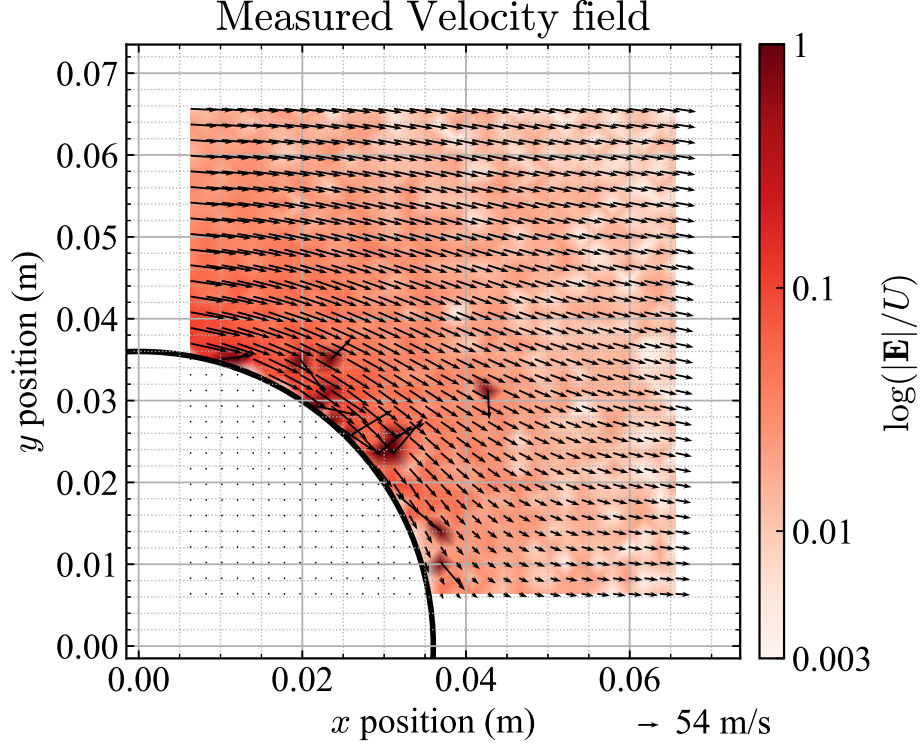


Figure 22: A simulated velocity field measurement for $Ut_\delta/d_c = 3 \times 10^{-2}$ and otherwise ‘typical’ system parameters, coloured to show the normalised magnitudes of measurement errors.

5.3.10 Sensitivity to Exposure-Interframing Time Ratio

Figure (23) is a 2D histogram showing the variation of dimensionless measurement error against the ratio of exposure time to interframing time. For exposure-interframing time ratios less than 1×10^{-3} the median normalised RMS error is small as most particles remain inside a single pixel during each exposure.

For exposure-interframing time ratios between 1×10^{-3} and 2×10^{-2} the median normalised RMS error becomes large. This is due to the exposure being just long enough for a significant number of particle images to be blurred across two pixels. Whether or not a particle crosses from one pixel to another during an exposure depends on the position of the particle at the beginning of the exposure. For exposure-interframing time ratios between 1×10^{-3} and 2×10^{-2} some particle image centres are blurred across two pixels in one image and across one pixel in the other image. This mismatch causes false correlation errors. This region of high median normalised RMS errors is due to some of the ‘typical’ system’s dimensionless parameters lying close to sharp increases in normalised RMS error, meaning the system is not particularly robust. If the system was moved away from these sharp error increases it would not have this region of large normalised RMS error. For example, figure (15) shows that the image density of the ‘typical system’ lies very close to a sharp increase in median normalised RMS error. Increasing image density to move away from this point would remove the region of frequent false correlation errors between

$t_e/t_\delta = 1 \times 10^{-3}$ and $t_e/t_\delta = 2 \times 10^{-2}$ seen in figure (23). Highlighting regions with sharp increases in normalised RMS error is therefore a useful application of the digital twin.

Between exposure-interframing time ratios of 2×10^{-2} and 8×10^{-2} the median normalised RMS error is small again, due to the exposure time being large enough for the blur of particles to be fairly consistent between the two images, reducing the frequency of false correlation errors.

Exposure-interframing time ratios above 8×10^{-2} introduce errors due to the blurring of particles making the correlation peak less sharp and therefore resulting in measurements which are slightly different to the velocity field. Above 2.6×10^{-1} the number of false correlation increases greatly.

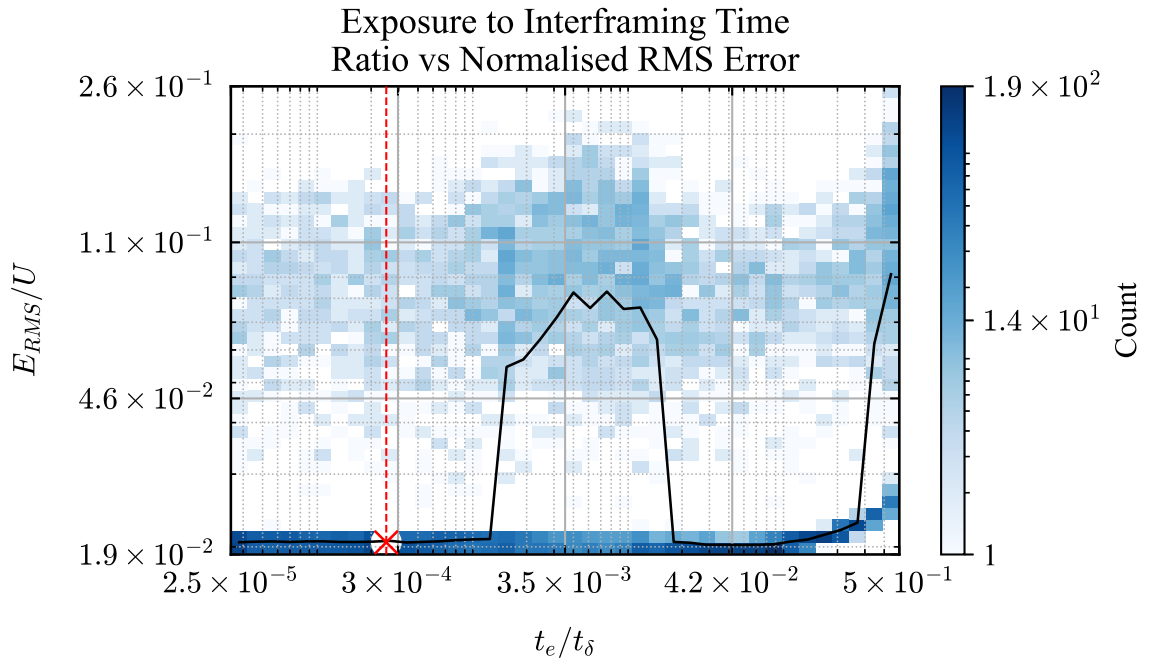


Figure 23: A 2D histogram and median trend line for the normalised RMS error against the ratio of exposure time to interframing time, for 8600 simulated PIV measurements at 86 values of the aforementioned ratio.

It is recommended that the ‘typical system’ is operated with an exposure-interframing times less than 1×10^{-3} to avoid false correlation or blurring errors. It is possible to operate a high accuracy system with exposure-interframing times between 2×10^{-2} and 8×10^{-2} , but this is not recommended due to the sensitivity of blur statistics to flow velocity.

5.3.11 Sensitivity to Dimensionless Particle Image Size

Figure (24) is a 2D histogram showing the variation of dimensionless measurement error against the ratio of the ‘particle image sigma’ to the interrogation spot width. The

‘particle image sigma’ is the standard deviation of the 2D Gaussian distribution governing the brightness imparted to each pixel by a single particle. Equation (5) defines how σ affects the simulated PIV images. Increasing ‘particle image sigma’ increases the ‘size’ of the image created by a single particle.

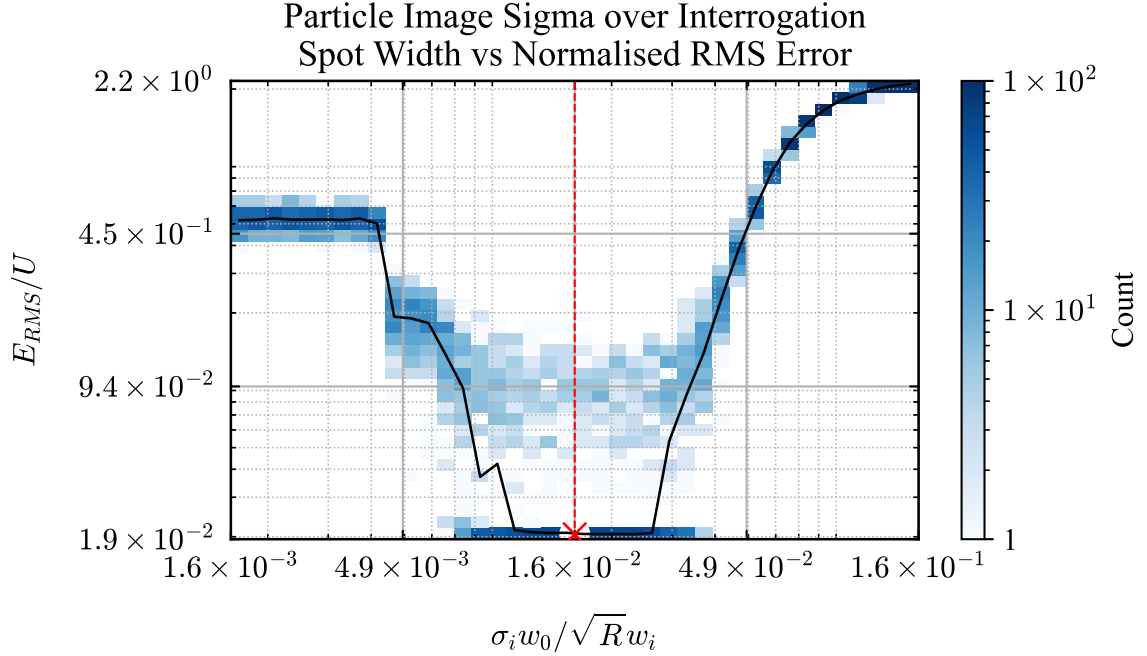


Figure 24: A 2D histogram and median trend line for normalised RMS error against the ratio of the standard deviation of the particle image brightness Gaussian distribution divided by the width of an interrogation spot, for 4000 simulated PIV measurements at 40 values of the aforementioned ratio.

The median normalised RMS error is large and constant for $\sigma_i w_0 / \sqrt{R} w_i < 4 \times 10^{-3}$ due to particle images occupying single pixels, as shown in figure (25a) for a particle image sigma to interrogation spot width ratio of 3×10^{-3} . Single pixel particle images result in frequent false correlation errors because shear or rotation causing a single pixel change in the position of a particle relative to the other particles in an interrogation spot is enough to cause a false correlation error. These false correlation errors are seen in figure (26) for a simulated velocity field measurement with $\sigma_i w_0 / \sqrt{R} w_i = 3 \times 10^{-3}$. The false correlation errors are concentrated in the region of high image shear distortion near the cylinder. See section 5.3.9 for a more detailed explanation of the effect of flow shear on errors.

The median normalised RMS error drops rapidly once $\sigma_i w_0 / \sqrt{R} w_i$ is large enough that particle images illuminate the four pixels adjacent to the central image pixel, as seen in figure (25b). Figure (25b) has had its brightness artificially increased after simulation to enable the reader to see the four illuminated pixels around each central pixel. For $1.2 \times 10^{-2} < \sigma_i w_0 / \sqrt{R} w_i < 2.6 \times 10^{-2}$ the median normalised RMS error is small.

Increasing $\sigma_i w_0 / \sqrt{R} w_i$ above 2.6×10^{-2} leads again to an increase in false correlation errors. However, the false correlation errors at large dimensionless particle image sizes are due to particles being indistinguishable from each other due to excessively large particle images. These false correlation errors are spread uniformly over the whole field of view, as seen in figure (27) which shows a simulated velocity field measurement for $\sigma_i w_0 / \sqrt{R} w_i = 5 \times 10^{-2}$.

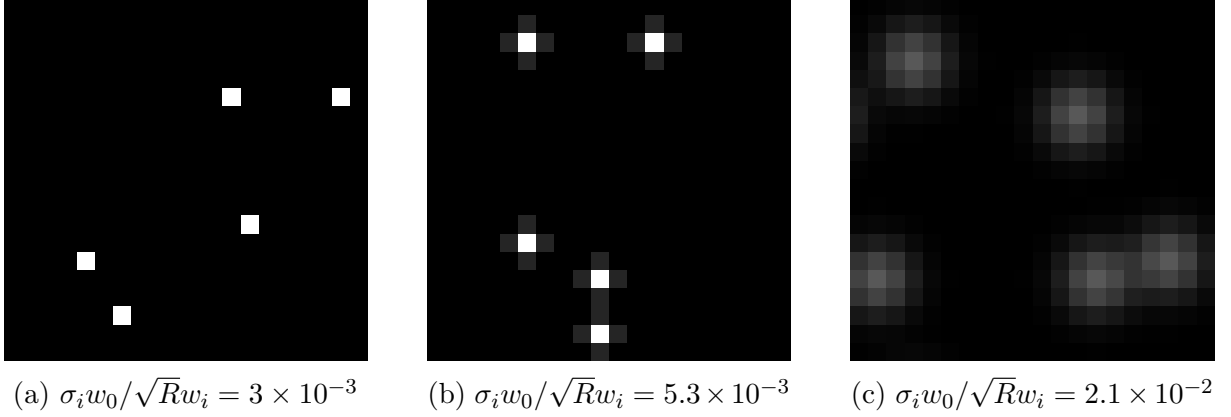


Figure 25: Three 20×20 pixel² sections of simulated PIV images from systems with varying dimensionless particle image size and otherwise ‘typical’ parameters as stated in table (2). Figure (25b) has been artificially brightened after image generation to allow the reader to see low brightness regions of the particle image.

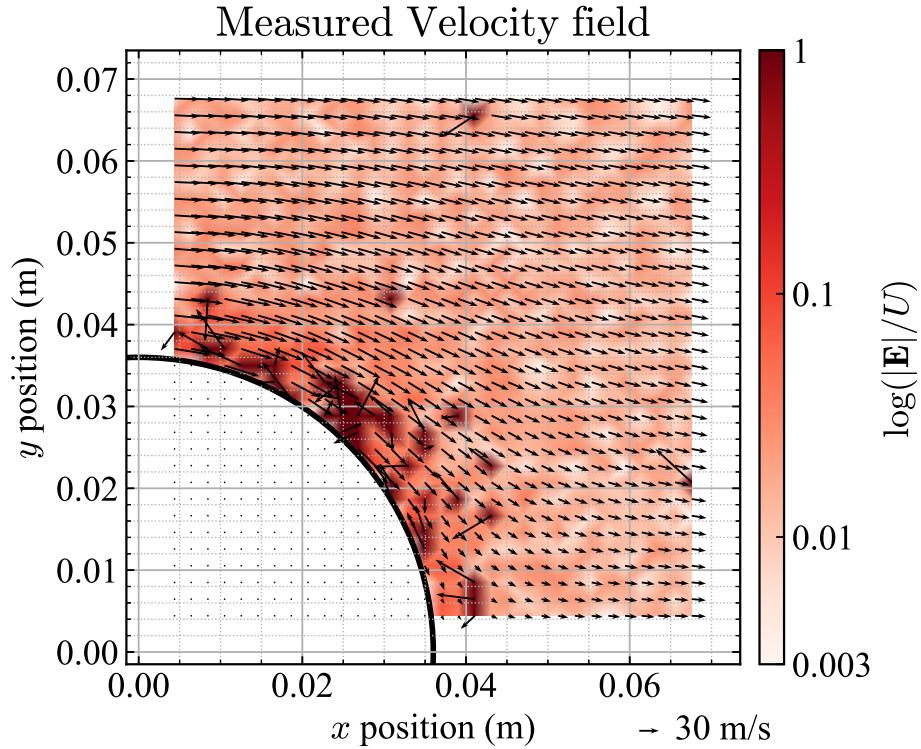


Figure 26: A simulated velocity field measurement for $\sigma_i w_0 / \sqrt{R} w_i = 3 \times 10^{-3}$, coloured to show the normalised magnitudes of measurement errors.

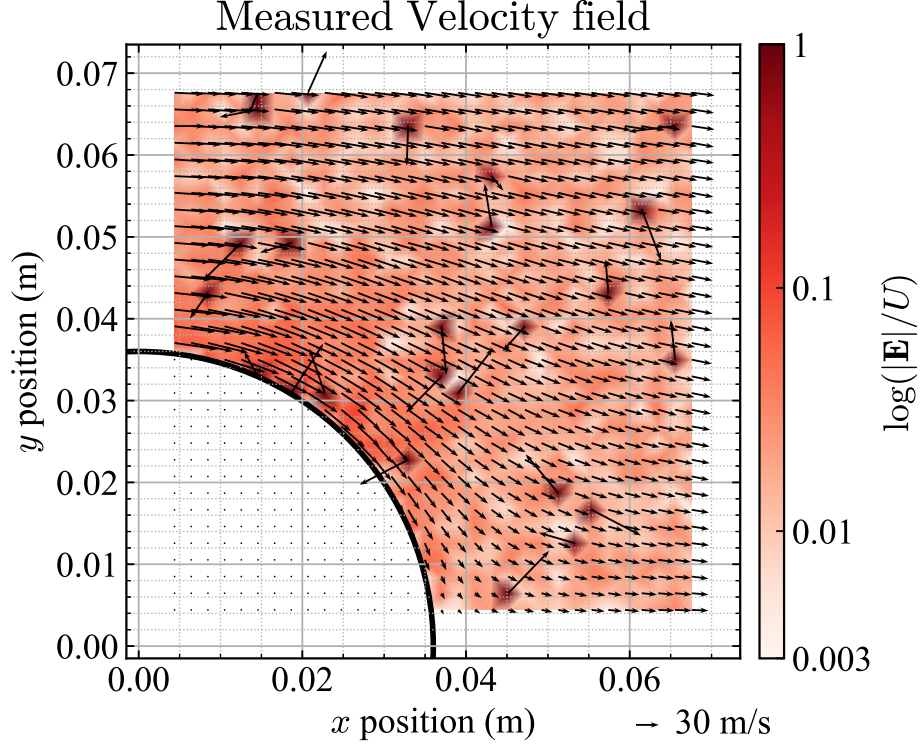


Figure 27: A simulated velocity field measurement for $\sigma_i w_0 / \sqrt{R} w_i = 5 \times 10^{-2}$, coloured to show the normalised magnitudes of measurement errors.

5.3.12 Sensitivity to Dimensionless Particle Brightness

Figure (28) is a 2D histogram showing the variation of dimensionless measurement error against characteristic particle image brightness. The dimensionless error is large and constant for values of characteristic particle image brightness $It_e/R < 1 \times 10^{-5}$ because the illumination is not sufficient to see the particles. For $1 \times 10^{-5} < It_e/R < 1.5 \times 10^{-4}$ the median normalised RMS error falls rapidly with increasing particle image brightness, as first the centre of each particle image becomes visible, and eventually the whole particle image becomes visible. Above $It_e/R = 1.5 \times 10^{-4}$ the median normalised RMS error magnitude remains small and constant. In practice illumination intensity is limited by the requirement that the illumination source does not vaporise the seeding particles.

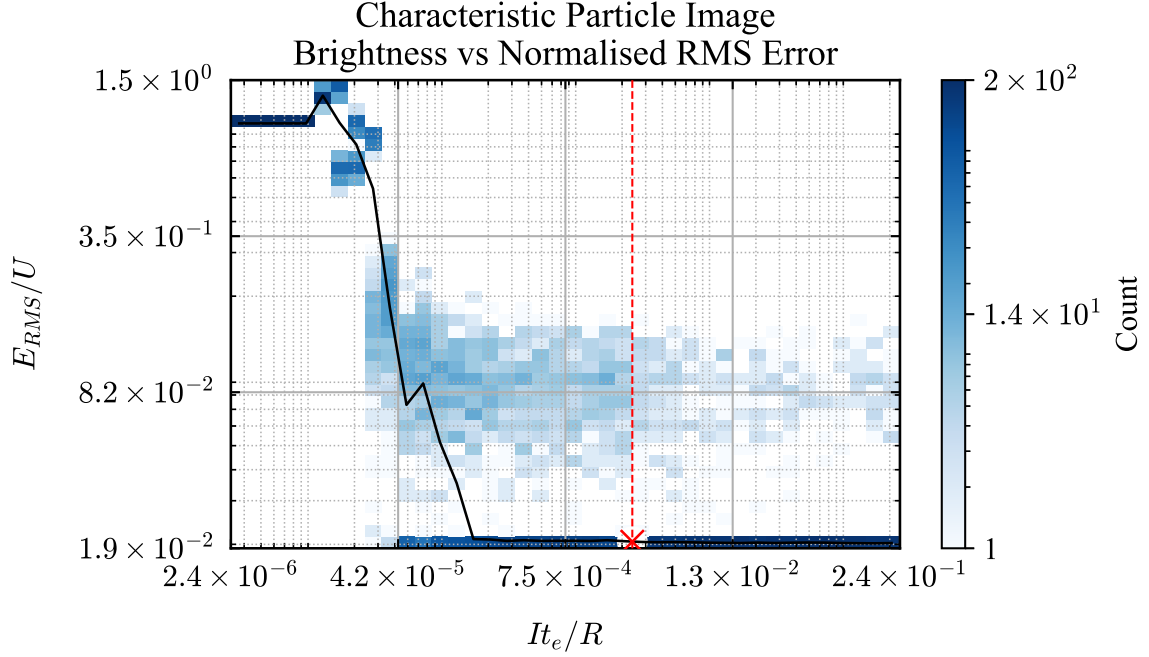


Figure 28: A 2D histogram and median trend line for the normalised RMS error against the characteristic particle image brightness, for 8000 simulated PIV measurements at 80 values of characteristic particle image brightness.

6 Case Studies

The digital twin is primarily useful for providing simulated data about PIV system errors. This allows systems with small errors to be designed and their limits understood before actually building the system. Three case studies are presented to show how design requirements can be converted into a PIV system concept using the digital twin. The first case study concerns the design of a low cost PIV system to measure the flow of water around a cylinder for a fluid dynamics teaching lab. The second case study details a novel dual camera PIV approach and its potential for measuring the flow around a cricket ball. The final case study explores the possibility of using a pressurised wind tunnel and Reynolds matching to achieve improved PIV accuracy with a commercial system.

6.1 PIV in Water

Consider designing a low cost PIV system for use in a fluid dynamics teaching lab. The system is required to measure the flow behind a 1 cm diameter cylinder immersed in water at 0.15 m/s. It should nominally have a median normalised RMS error less than 5%, a field of view of 1 cm² (to inspect the flow at a specific location in the cylinder wake) and should be designed using off the shelf parts.

A system was designed to this specification using the Allied Vision Alvium 1800 U-

052m camera, costing £413, ‘Conduct-o-fil’ hollow ceramic spheres, which cost about £15 for enough to perform a PIV experiment (based on Acquah et al.[4] and converting currencies), and an LPH1-04C24-2780-00 100W LED module as the illumination source, costing £40.85. Table (4) details the properties used to simulate this system in the digital twin. The particle properties are the nominal average particle properties from the Conduct-o-fil website, the flow and field of view properties are chosen to represent an arbitrary lab experiment and the interrogation properties are chosen for low system errors. The exposure time is a camera property in this system due to the continuous illumination source, and the interframing time is set to the time between frames at the camera’s maximum frame rate. The number of particles can be varied in a real experiment and hence is set to a value which produces low system errors. The particle image size and illumination intensity are estimates based on scaling from the ideal system illumination intensity. Scattered light energy scales with the square of particle size for particles larger than $1\mu\text{m}$ [2], so for a system with particles approximately 300 times larger than the typical system and with illumination intensity roughly five orders of magnitude lower than the typical system (based on the difference in luminous energy between a pulsed laser source and an LED illumination source based on the hardware review in section 4.1) the intensity parameter I remains the same as for the typical system.

Conduct-o-fil AG-SL150-30-TRD	d_p	$100 \times 10^{-6} \text{ m}$
	ρ_p	1100 kg m^{-3}
Measuring the flow of water around a cylinder for a teaching lab	U	0.15 m s^{-1}
	d_c	$1 \times 10^{-2} \text{ m}$
	μ	$8.9 \times 10^{-4} \text{ kg m}^{-1} \text{ s}^{-1}$
	ρ_f	1000 kg m^{-3}
	w_0	$1 \times 10^{-2} \text{ m}$
Allied Vision Alvium 1800 U-052m	R	509184 pixels^2
	t_e	$21 \times 10^{-6} \text{ s}$
	t_δ	$1.447 \times 10^{-3} \text{ s}$
Chosen Interrogation Properties	w_i	$6.25 \times 10^{-4} \text{ m}$
	N_o	1
Estimated Properties	n_p	5120
	I	$1 \times 10^{12} \text{ pixels}^2 \text{ s}^{-1}$
	σ_i	1 pixel

Table 4: System parameters for a PIV system designed to measure the flow of water behind a cylinder in a lab experiment

The system in table (4) produces a median normalised RMS error of 4.9%, satisfying the design requirements. Therefore, this is a sensible choice of camera, tracer particles, and illumination. The main three components (the camera, illumination, and seeding particles) cost a total of £469. This example demonstrates how the digital twin can be used with real parameters to provide information about system errors before a system is

constructed.

6.2 Low Cost Cricket Ball PIV

Consider designing a low cost PIV system for use in an aerodynamics laboratory, designed to measure part of the flow around a cricket ball at 30 m/s. This system is required to measure a 25 cm² region of the flow around the cricket ball, which has a diameter of 72 mm, and the main three components (the camera, illumination, and seeding particles) should cost under £1000.

CMOS cameras with large enough frame rates and resolutions to perform PIV for the flow around a cricket ball at this speed do not exist within the stated price range. A novel idea to circumvent this issue is to use two CMOS cameras positioned on either side of the wind tunnel and carefully calibrated to capture the same fields of view. The interframing times required for cricket ball PIV could then be achieved by using an electronic camera trigger and a built in delay in one of the cameras. This setup is investigated for two FLIR Blackfly S GigE cameras, costing £346.50 each. It is assumed that the cameras can be aligned sufficiently accurately.

Table (5) shows the properties used to simulate this system. The flow around a cricket ball is different to potential flow around a cylinder, but the velocities will have the same order of magnitude, so the digital twin should give a reasonable indication of the expected error magnitudes. 1 µm atomised vegetable oil was chosen as the seeding particle, as a seeding nozzle can be manufactured by the aerodynamics laboratory and a bottle of vegetable oil costs around £2. Two LPH1-04C24-2780-00 100W LED modules, costing a total of £81.70, are chosen for the illumination. The illumination intensity parameter is estimated by scaling the value for the ‘typical system’ by a factor of 5×10^{-4} for the approximate change in illumination intensity compared to a pulsed laser source, and by a factor of 10 corresponding the diameter increase of the particle squared[2].

The system detailed in table (5) achieves a median normalised RMS error of 9.0%. The cost of the camera, illumination, and seeding system totals £777, excluding the cost of nozzle manufacture which is assumed to be free within the aerodynamics department. Therefore this system meets the specified requirements within the budget and the hardware choices are appropriate, provided the camera field of views can be calibrated sufficiently accurately. Again, the digital twin has provided error magnitude information for a hypothetical system, demonstrating its usefulness for PIV system design.

Atomised Vegetable Oil	d_p	$1 \times 10^{-6} \text{ m}$
	ρ_p	920 kg m^{-3}
Air flow around a cricket ball	U	30 m s^{-1}
	d_c	$7.2 \times 10^{-2} \text{ m}$
	μ	$18.5 \times 10^{-6} \text{ kg m}^{-1} \text{ s}^{-1}$
	ρ_f	1.2 kg m^{-3}
	w_0	$5 \times 10^{-2} \text{ m}$
	R	345600 pixels^2
Two FLIR Blackfly S GigE cameras	t_e	$20 \times 10^{-6} \text{ s}$
	t_δ	$4 \times 10^{-5} \text{ s}$
	w_i	$4.17 \times 10^{-3} \text{ m}$
Chosen Interrogation Properties	N_o	1
	n_p	2880
Estimated Properties	I	$2 \times 10^8 \text{ pixels}^2 \text{ s}^{-1}$
	σ_i	1 pixel

Table 5: System parameters for a PIV system designed to measure the flow of air around a cricket ball using two cameras

6.3 High Pressure Reynolds Matched PIV

Designing a PIV system is easier for lower speed flows due to the longer permissible exposure and interframing times. Choosing seeding particles which accurately follow the flow is easier with higher density fluids. This leads to the idea of increasing wind tunnel pressure and matching Reynolds number in order to perform PIV on a system with increased fluid density and decreased fluid speed.

Consider using a PIV system to measure the boundary layer flow for an aerofoil designed to operate at 100 m/s. Assume we measure this aerofoil at a point with an instantaneous radius of curvature of 0.25 m. The parameters used to simulate a commercial PIV system using the ‘pco.panda DS’ camera (£7800), the ‘BeamQ 527 nm Double Pulsed Laser’ (£6610), and atomised oil as seeding particles (negligible cost if atomising nozzle is manufactured in house) are shown in table (6). The system is designed to measure a 4 cm² region of the flow at 1600 interrogation spots. The oil particles are assumed to have the same diameter as in the ‘typical system’ and the parameters not stipulated by the camera, laser, or particles are set to produce small errors.

Simulating both the system measuring the flow at atmospheric pressure and the Reynolds matched flow pressurised to 10 bar leads to median normalised RMS error magnitudes of 7.8% and 1.6% respectively. This discrepancy is due to the change in Stokes number affecting the accuracy with which the tracer particles follow the flow. These simulated results demonstrate the efficacy of pressurised Reynolds matched PIV as a tool to measure small regions of high speed flows. This example again demonstrates the usefulness of the digital twin as a tool to provide information on PIV system error magnitudes instead of relying on costly experimentation.

PIV System	Variable	$p = 1 \text{ bar}$	$p = 10 \text{ bar}$
Atomised Oil	d_p	$0.3 \times 10^{-6} \text{ m}$	$0.3 \times 10^{-6} \text{ m}$
	ρ_p	920 kg m^{-3}	920 kg m^{-3}
Boundary layer flow around an aerofoil with a design speed of 100 m/s	U	100 m s^{-1}	10 m s^{-1}
	d_c	0.5 m	0.5 m
	μ	$18.5 \times 10^{-6} \text{ kg m}^{-1} \text{ s}^{-1}$	$18.5 \times 10^{-6} \text{ kg m}^{-1} \text{ s}^{-1}$
	ρ_f	1.2 kg m^{-3}	12 kg m^{-3}
	w_0	$2 \times 10^{-2} \text{ m}$	$2 \times 10^{-2} \text{ m}$
pco.panda DS	R	$26214400 \text{ pixels}^2$	$26214400 \text{ pixels}^2$
Camera and BeamQ	t_e	$2 \times 10^{-7} \text{ s}$	$2 \times 10^{-7} \text{ s}$
Double pulsed laser	t_δ	$2 \times 10^{-6} \text{ s}$	$20 \times 10^{-6} \text{ s}$
Chosen Interrogation Properties	w_i	$5 \times 10^{-4} \text{ m}$	$5 \times 10^{-4} \text{ m}$
	N_o	1	1
Estimated Properties	n_p	32000	32000
	I	$2 \times 10^{10} \text{ pixels}^2 \text{ s}^{-1}$	$2 \times 10^{10} \text{ pixels}^2 \text{ s}^{-1}$
	σ_i	1.5 pixel	1.5 pixel

Table 6: System parameters for a PIV system designed to measure the boundary layer flow around an aerofoil at a region with an instantaneous radius of curvature of 0.25 m. The aerofoil is designed to operate at 100 m/s, and the table shows the simulated parameters for a PIV measurement at the design conditions, and for a PIV measurement in a pressurised wind tunnel with conditions set to match Reynolds number.

7 Conclusions

No tools exist capable of performing all of the following tasks: synthetic PIV image generation including pixelisation, particle simulation and optical modelling; velocity estimation using synthetic or real PIV images; and comparison of simulated velocity measurements to an exact underlying flow. The ‘digital twin’ was developed to fill this gap and enable the errors associated with any PIV system to be quantified before system construction.

Dimensional analysis shows the median dimensionless measurement error for a simulated system depends on twelve dimensionless parameters, and the sensitivity of median dimensionless error for a ‘typical system’ to variations in dimensionless parameters was analysed using the digital twin. The ratio of particle density to fluid density, Reynolds number, and interrogation spot overlap factor do not affect median dimensionless error. The numerical values presented in the following sensitivity analysis discussion pertain to variations of a single dimensionless variable about the ‘typical system’ only.

Pixel densities below 1.8×10^4 cause excessive false correlation errors due to particle groups being indistinguishable. Quantisation errors dominate above 1.8×10^4 and decrease with increasing pixel density as the resolution of velocity measurement increases.

Dimensionless errors are small for Stokes numbers below 5×10^{-5} and increase with Stokes number above this value due to particle slip. The median dimensionless error magnitude is acceptable for Stokes numbers below 1×10^{-4} .

Decreasing the cylinder diameter to field of view width ratio below 1 causes increasing pixelisation errors, while increasing this ratio above 1.25 causes a sharp increase in false correlation errors. Median dimensionless error magnitude is small for values of this ratio between 1 and 1.25. This range may be expanded by increasing pixel density.

Median dimensionless error is small for image densities between 23 and 310. Below this range false correlation errors occur as there are too few particles to confidently correlate all interrogation spots, and above this range false correlation errors occur due to dense groups of particles being indistinguishable. Increasing the ‘typical’ system’s image density from 12 to a value between 23 and 310 would decrease median dimensionless error and improve system robustness to other dimensionless parameter variations.

Interrogation spot to FOV width ratios smaller than 2.7×10^{-2} yield false correlation errors due to interrogation spots being small relative to the extents searched for correlated particles. Median dimensionless error is minimised at an interrogation spot to FOV ratio of 2.7×10^{-2} and above this value dimensionless error increases due to increasing velocity variation within the increasingly large interrogation spots.

Decreasing the ratio of freestream interframing distance to cylinder diameter causes increasing pixelisation errors due to small particle travel distances, while increasing the ratio above 2×10^{-2} causes false correlation errors due to large shearing motion in the flow. Freestream interframing distance to cylinder diameter ratios between 1×10^{-2} and 2×10^{-2} give small median dimensionless error. This region of low dimensionless error can be expanded by increasing pixel density.

Increasing the exposure-interframing time ratio above 8×10^{-2} increases dimensionless error because excessively long particle streaks flatten the true-shift correlation peak. Ratios between 1×10^{-3} and 2×10^{-2} generate many false correlation errors which could be mitigated by increasing the number of tracer particles. Ratios below 1×10^{-3} produce a small median dimensionless error.

Dimensionless particle image sizes between 1.2×10^{-2} and 2.6×10^{-2} produce small dimensionless errors. Below this region particle images are small and shear causes false correlation errors. Above this region particle image overlap causes false correlation errors.

Characteristic particle image brightness values below 1×10^{-5} fail to illuminate the particles, causing large errors. Increasing brightness above this value rapidly reduces the frequency of false correlation errors, with small and approximately constant median dimensionless error for characteristic brightness values above 1.5×10^{-4} .

A hardware review revealed that the pulsed lasers used in conventional PIV systems offer instantaneous luminous power ratings five orders of magnitude greater than off the shelf LEDs, but LEDs are around three orders of magnitude cheaper. The CCD cameras used in conventional PIV are more expensive than most CMOS cameras, but CCD cameras’ interframing times are around three orders of magnitude smaller.

Case studies were performed, using the digital twin to assist conceptual design of a

water based PIV system and a novel dual-camera low cost PIV system. The digital twin was also used to show how increasing air pressure and Reynolds matching can reduce experimental uncertainty by reducing particle slip errors.

Future work includes extending the digital twin to link the illumination model to hardware parameters, and testing the low cost PIV systems and experimental methods proposed in the case studies. Additionally, implementing different underlying flow fields and correlation methods with sub-pixel accuracy would improve the digital twin's ability to model real PIV measurements made by state of the art systems.

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9 Risk Assessment Retrospective

The only risks encountered in this project were computer related repetitive strain injuries. Standard techniques such as maintaining good posture and taking regular breaks were used to effectively mitigate all risks.